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2020-2021 Winter Edition

The Official Mathematics and Computer Science Publication
of the Stuyvesant High School Math Department since 1927

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Visit <https://stuymathsurvey.com> for the online version and for interactive visualizations.



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Editor's Note

Josephine Lee

[W]e are not mathematical machines. We live, we breathe, we feel, we bleed. We are embodied human beings. Why should anyone learn mathematics if it doesn't connect deeply to some human desire, something we long for - to play, seek truth, pursue beauty, find meaning, fight for justice? You, as a mathematical explorer who is learning to embrace your identity in math, can be part of the movement to read people differently.

- Francis Su, *Mathematics for Human Flourishing* (2020)

How would you personally answer Professor Su's question? Why should you learn mathematics? In his compelling book published in early 2020, *Mathematics For Human Flourishing*, Professor Su gives us an in-depth look into 12 intriguing reasons to understand and to contribute to the glorious field of mathematics: exploration, meaning, play, beauty, permanence, truth, struggle, power, justice, freedom, community, and love.

I am still formulating my answer to his probing question, but the events of 2020 and 2021 have shaken us to the core of our beings and have made us reflect on fundamental problems in society into which math can provide insight. The devastation caused by the global pandemic, heightened tension in racial relations, and intense political strife has forced us to think more deeply about what pursuits are worthy of our time, treasure, and talents.

I was motivated to revive *Math Survey* in the fall of 2019 by my desire to creatively engage further with Stuyvesant's diverse and spirited math community. Little did I know that the crises in 2020 would make human exploration and connection absolutely paramount.

Despite all the recent monumental stresses on the Stuyvesant administration, faculty, staff, and students, the re-publication of *Math Survey* became a reality because of the dedication demonstrated by so many. Each of them deserves our sincere gratitude.

Principal Yu, thank you so much for stepping in this year and steering Stuyvesant through these unprecedented times. I'm looking forward to meeting you in person someday!

Principal Contreras, we appreciate how welcoming and encouraging you were during your years at Stuyvesant. Every time I walked through the door of Room 105, you showed me genuine interest and warmth.

Mr. Smith, I am grateful that you have been open-minded and responsive each time I've met with you and that you lead the math and computer science department so thoughtfully.

Mr. Rubinstein, thank you for believing in me and enthusiastically guiding me every step of the way as our faculty advisor, ever since my very first email to you. Your dedication to teaching and to math and computer science inspires us all. *Math Survey* is forever in your debt!

To the Stuyvesant Parents' Association, we greatly appreciate your support for the printing of *Math Survey* and enabling us to take great strides in publishing the inaugural Winter Edition.

Last, but certainly not least, to each member of our hard-working *Math Survey* staff, thank you for the tremendous effort you have put into meaningfully contributing to this enduring publication that started in 1927 with like-minded students. You are supporting the entire Stuyvesant math and computer science community in its commitment to human flourishing.

Foreword

Math Research - A Marathon, an Obsession, and a Beautiful Picture

Professor Michael Hutchings (Class of 1989) is a mathematician and a professor at University of California, Berkeley. He attended Stuyvesant from 1986 to 1989, where he contributed to the Math Team and Math Survey, and Harvard College from 1989 to 1993. As an undergraduate student at Harvard College, Hutchings did a Research Experience for Undergraduates (REU) project with Frank Morgan at Williams College that launched his interest in the mathematics of soap bubbles. After postdoctoral and visiting positions at Stanford University, the Max Planck Institute for Mathematics in Bonn, Germany, and the Institute for Advanced Study in Princeton, New Jersey, he joined the University of California, Berkeley faculty in 2001. He is currently the Chair of the Math Department at University of California, Berkeley and is known for the development of embedded contact homology and for proving the double bubble conjecture on the shape of two-chambered soap bubbles.

This foreword is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

My math journey started when I entered Stuyvesant as a sophomore after I moved to New York City. One day, I just wandered into the Math Team, which had practice every day at 8:00 AM, where they were doing a practice test that day. The captain was Elizabeth Wilmer, who also later became a math professor, and the coach was Mr. Geller. They, along with some of the other older students in the Math Team, encouraged me and were essential mentors and teachers for me. A lot of what made Stuyvesant a good school for me was the fellow students. The Math Team connected into this whole circle of people and groups and activities. I went to the Hampshire College Summer Studies in Mathematics (HCSSiM), which I wouldn't have known about if I hadn't learned about it from the Math Team. Before that summer camp, I wasn't sure that I wanted to be a mathematician because I didn't really have much of an idea of what a mathematician did. I initially thought a mathematician would compile tables of integrals, or try to compute more digits of Pi, which didn't sound terribly interesting to me. HCSSiM introduced more of what math research is like, which is completely different from competitions. I got at least a little glimpse of how there's this whole huge world of interesting math problems and cool,

creative ideas that people are working on. I also wrote some articles for the *Math Survey* back then about tiling with quadrilaterals and how to do mental arithmetic quickly.

When I started my career in math, I had a very skewed view of things because the only really interesting math available to me was the Math Team. Math competitions were all about solving problems quickly. You have to find the clever trick, or maybe even read the mind of the person who wrote the question. It's like a sprint, but research is more like a marathon. It could take years to solve a problem. Collaboration is extremely important. Research is more about being persistent and working on complicated puzzles over a long period of time and collaborating and exchanging ideas with other people. You share ideas with other people because maybe I'm an expert on one thing but somebody else is an expert on something else, and by combining our knowledge we can do something really cool. I find that in research, if you're in the same room with your collaborators working on a problem together, it's sometimes really magical and you can do things which you never anticipated would be possible. When you finally figure the problem out, it's very satisfying. You usually find some beautiful picture.

As for the subject matter, my research area is in geometry and topology. This has little resemblance to the geometry that one usually sees in high school and on math competitions. The latter tends to focus on Euclidean geometry, and somewhat obscure questions about triangles and circles. I was actually never very good at that in math competitions. What modern geometers do is much more interesting. It's about curved spaces in three or four dimensions or higher, such as the shape of the whole universe, black holes, soap bubbles, or knots, to name a few. There are also some cool things about being a mathematician - there's tons of travel to participate in conferences and work with collaborators, which I had no idea about. I've spent a total of years of my life in other countries for work, especially in Germany, Brazil, and East Asia. I'm very lucky that I get to work on cool puzzles as part of my job.

The first time that I really failed badly at something was in the middle of graduate school. I spent a year trying to solve a certain math problem, and I thought I had a good idea for how to solve this problem. I thought I figured it out and I worked hard to write a research paper explaining my solution to the problem. Then it turned out, a couple of days after I finished the paper, I discovered there is a serious error in there and no apparent way to fix it. So this whole year of work was a total failure. This happens of course; it's a part of life as you try hard things. You're going to have to fail some of the time. The younger you can have this experience and get over it the better because it gets harder. In my case, I was very discouraged and I was considering quitting graduate school and getting a computer programming job. But I pulled myself together and found some other good stuff to work on with the help of my advisor, which eventually led to my PhD thesis.

Working on a problem for years requires a certain kind of personality. You have to be obsessed. Now I should clarify that it's not that you pick one problem. Usually, you have various problems that you're working on. It's hard to isolate good questions that you can hope to answer. Some of

them might be impossible, or maybe you'll never figure them out and somebody else will later. And that's okay. You don't have to solve everything, and often you also don't really know what exactly the question is. In a math homework assignment or on a math contest, you're given a question and you answer it if you can, but in math research, many times you don't even know what the question is. It also takes a really long time to write your research papers. If all I had to do was sit and write my research papers all day, I would be very happy. Unfortunately, this is unfeasible.

In terms of my research, my biggest accomplishment is the development of a theory in three- and four-dimensional geometry and topology called "embedded contact homology". In retrospect, this was a big gamble, because I worked for years on developing this theory, without knowing in advance whether it would be successful or have any value. I wasn't really thinking of it like that, though; I was just working on it because I was interested in it. In any case, it turned out to be successful and useful for some problems in geometry, specifically symplectic embeddings (a kind of packing problem), and dynamics of Reeb vector fields (roughly speaking, understanding behavior of physical systems that repeats in time). Aside from doing research, I have taught many thousands of students in my classes over my whole career. A lot of the students probably found my class boring, and maybe didn't get that much out of it. But just given such a large number of students, if I helped some of them learn to think more logically about quantitative questions, that might make more of an impact on the world than any of my research.

During the pandemic, a few things in regards to math research have arguably gotten better because there are research seminars and conferences that are now held online. Anybody from anywhere in the world can come and learn. In regards to the future of mathematics, there is some talk about proof assistants, where you have some software where you can write your mathematical proof with computer code, and the proof assistant can verify that your proof is correct. The proof assistant can even fill in some small details, so you don't necessarily have to write down every single step. However, I'm not sure that's going to happen that much with the kind of geometry that I do; it's very difficult to translate geometrical reasoning into computer code. And even if you do, there's a problem making sure that the interpretation of your computer code is actually what you were trying to say. So in other words, a lot of the difficulty gets shifted onto making sure that your definitions are really defining what you mean to be defining. So while that's an interesting development, I think it will be a while before it has a big impact on the practice of mathematics. And even if a computer can check that your proof is correct, it seems unlikely that computers can replace the human intuition that led to the proof in the first place.

There is also room for more computer experimentation in mathematics, which seems to be underutilized at least in the kind of geometry that I work on. There will most likely be more potential for students to go into mathematics with good computer programming skills that can be used to do experiments, to test mathematical conjectures.

My advice for those of you aspiring to go into math would be to follow what you're interested in, talk to other people as much as you can, and learn from other people. In the next 10 or 15 years, what being a mathematician means most likely won't change too much. Of course, there's very rapid progress in mathematics - new fields of mathematics emerge, new things become possible, and mathematical advances that would have been inconceivable 10 or 20 years previously become possible. I can't predict what's going to happen in terms of the content of mathematics. In my own specialty, there are some famous conjectures; personally I am most obsessed with one called the Viterbo conjecture. I try to make little bits of progress related to some of these conjectures. So maybe we can solve some of these, and I am certainly very excited to see what will happen. But in many ways, the practice of mathematics research will probably stay the same. There's not that much to it. It's reading papers, thinking, talking to people, and writing papers. And I think we're going to keep doing that.

Michael Hutchings
December 2020

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Crinkler: The Real-World Use of an Esoteric Compression Algorithm

Victor Veytsman

1 Introduction

Often-times in the Computer Science world, a brilliant paper will be published, an excellent algorithm made that elegantly solves a problem in a way never seen before, only to never be used, for practical reasons. It might take up too much space, or be too slow, or be too hard to implement, or only work well for an edge case too specific to consider. Whatever the reason, there is often a disconnect between the cutting edge of Computer Science and what people end up using on a day-to-day basis. This article delves into software called *Crinkler*, which demonstrates how algorithms that would otherwise be restricted to a few scientific papers buried in some university's archive can be used in the real world. But before we take a look at Crinkler, we must examine what spawned the software in the first place: the *Demoscene*.

1.1 What is the Demoscene?

The Demoscene is a fascinating subsection of Internet culture that brings people together to push their code to its limits. “Sceners,” the demonym for those who participate in this culture, create “demos” which assemble impressive music and graphics into programs that perform awe-inspiring stunts. There are many different types of demos, most putting a unique constraint on the demo that works to highlight the scener's creativity. The most common category is that of “intros,” which are restricted by their size, which is the amount of disk space they can take up. Other categories include different retro platforms, e.g., computers like the Amiga or the Commodore 64, where their low level of performance provides the restrictions.

The Demoscene has its roots in the early 1980s when computers first started entering the home. Games released for these computers had copy protection, to make sure people didn't spread them illegally, so lots of people, calling themselves “crackers,” worked on breaking this protection to share the game for free. When they cracked a game, they would often add an introduction to it that shouted them out. As the 80s went on, crackers would keep trying to one-up each other, fitting impressive graphics into the little space they had. They also met each other at parties to swap their cracked games.

Nowadays, the Demoscene lives primarily in Europe, having moved away from cracked games to original creations. Demoparties are hosted in countries like Germany, Switzerland, Denmark, Norway, and Sweden, and many Internet forums are dedicated to hosting demos and discussions about them.

2 What is Crinkler? How does it work?

To pull off their impressive stunts, demos employ a variety of tricks to make their code smaller and more efficient. A lot of demos rely on what are called “demotools,” which are tools created to make common

tasks easier and better. One of these very common tasks is compression, which is essential in the creation of intros, so there are many demotools whose primary purposes are compressing the code as efficiently as possible. Many of these tools fall into the category called “packers,” because they compress the demo and then add a small program onto it, called an “unpacker.” When the program is run, the unpacker runs first and decompresses the main demo.

Crinkler is one of these packers, made specifically for 4-kilobyte intros, programs that can only contain 32,000 zeros and ones (an amount of data equivalent to about one page of text). Crinkler is special, though, because it also acts as a *linker*. When you have code among a bunch of different files, the compiler generates a different *object file* for each code file. The object file is mostly machine code, code that the computer can read, but it also has some information for the linker to use. The linker then takes all of these files and puts them together into one file that the computer can run. Because Crinkler is also a linker, it can combine these object files in a way that makes them compress more efficiently, putting similar parts closer to each other and splitting up the code and the data according to their different structures. Once Crinkler finishes the linking process, it begins compression, taking advantage of several different algorithms.

2.1 Arithmetic Coding

Crinkler uses a technique called *arithmetic coding* to compress the file. Arithmetic coding is a compression algorithm that compresses the given data into a single number based on the probability of each symbol in that data occurring after another symbol. Crinkler makes this easier by going down to the bit level—it works with zeros and ones, whereas other implementations may work with bytes (eight zeros and ones) or even bigger blocks. To demonstrate how this algorithm works, here is an example:

Suppose there is a sequence of bits that is “0110.” Let’s say the probability table of this example is 0.6 for a zero and 0.4 for a one—it doesn’t matter whether or not this is perfect for the given sequence, and the creation of these probability tables is discussed later. The first step the algorithm takes is dividing the interval between 0 and 1 into smaller sections based on those probabilities. In Figure 1, the numbers on the left represent the interval between 0 and 1, and on the inside of each box is the sequence that the box’s interval describes. Since our sequence starts with a 0, we split our interval once again on the 0, as can be seen in Figure 2. Now, we have an interval between 0 and 1 that represents the sequence “01”, which is the start of our sequence, so we want to split that interval again. In Figure 3, this process has been completed twice to bring us to the final interval that we need to describe our full sequence, which is between 0.4 and 0.4576.

The algorithm can output any number that is within the interval. However, it would be most advantageous to output the shortest possible binary number that is within that interval. Finding this binary number is left as an exercise to the reader.

Now that we have the encoding done, we need a program to decode back to our original input. This is the beauty of arithmetic coding: the decoder is extremely simple, especially when you’re only dealing with zeros or ones. In essence, it does the same thing that the encoder does, but backwards. Given the encoded number, it divides the range between 0 and 1 into two sections. Then, it determines the section of the encoded number and outputs the bit corresponding to that section. Finally, it subdivides that section into two sections again, repeating the process for the length of the compressed code. When executing programs, the computer reduces them to instructions, where each instruction is then executed by the CPU. The developers of Crinkler have managed to get their decoder program to run with only 18 instructions,

each instruction not more than a few bytes (most being only one byte).

Figure 1: The first division of the interval

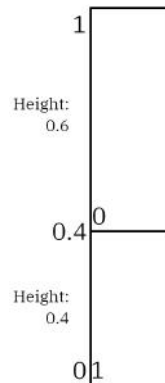


Figure 2: The second division of the interval

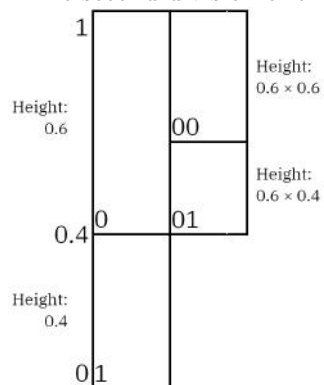
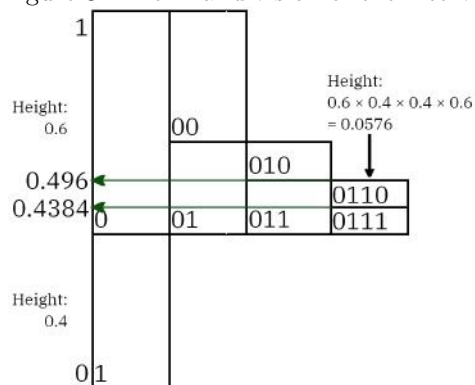


Figure 3: The final division of the interval



2.2 Making Probability Models

2.2.1 Making one Probability Model

With our previous example, a probability table said that there would be a 60% chance a 0 would occur and a 40% chance a 1 would occur in the sequence. These numbers were chosen as an example, but present an interesting question: how does the program make the probability models as accurate as possible? More accurate probability models should generate bigger intervals, which would make it easier to find a short number in that interval.

To create a model, Crinkler reads through the data, keeping track of the probabilities that bits occur and updating them as it reads new data. It stores a model for each context, the context being the previous bits it has read. Each model consists of two counters which count the number of times zeros and ones occur after their contexts. The algorithm slightly changes these counters in order to prioritize recent events over older ones. It does this by seeing if the opposite counter (the one whose bit didn't come after the model's context in the set the algorithm was reading) is greater than two, and if so, it divides it by two and adds one.

For example, let's take the sequence 010101100. Let's say that we have one model for this sequence whose historical data is 1 bit long. This means that the model is calculated twice: once for the historical data "0" and once for the data "1." The algorithm would keep track of all of the counters for each historical possibility at once, but for simplicity's sake, let's calculate just the model for each 0.

1. In the first pair of bits, 01, we add 1 to the counter: ($c_0 = 0, c_1 = 1$)
2. In the second pair of bits, we see the same thing: ($c_0 = 0, c_1 = 2$)
3. In the third pair, again: ($c_0 = 0, c_1 = 3$)
4. In the fourth pair, we see a 00, so we add 1 to c_0 , and then manipulate the counter as previously mentioned, rounding down: ($c_0 = 1, c_1 = \frac{3}{2} + 1 = 2$)

Now that we have calculated the counters for the 0 model, we can find the probability that a 1 occurs with this formula: $p(1) = \frac{c_1}{c_0 + c_1}$. Of course, it is impossible for $p(1)$ to be 0 or 1, so often a small number will be added to each counter to make sure that this impossibility never occurs; this is called a bland model. Models are added by simply adding their counter values, so adding our calculated model and our bland model (in which c_0 is 1 and c_1 is 1) gives us $p(1) = \frac{2+1}{(1+1)+(2+1)} = 0.6$. We estimated a 60% chance that after every 0 there will be a 1.

2.2.2 Making Many Probability Models and Combining Them

This way, Crinkler can calculate the probability of a zero or one occurring. However, it takes this whole procedure one step further by calculating 255 models. Each model is assigned a number from 0 to 255, which is 0b11111111 in binary. (The prefix 0b means that it's a binary number.) These numbers act as a *bit mask*; when converted to binary, the ones indicate which bytes are part of the historical data. This is seen more clearly in Figure 4 where each box represents a byte, with the green box indicating the one currently being read. With the zero model, only that byte is considered in the context; with the 128 model, the most recent byte is considered in the context (the binary number is backwards); with the 129 model, it's the 8th most recent and most recent bytes being part of the historical data; and with the 255 model, all of the most recent 8 bytes are part of the historical data. The reason Crinkler does this is compiled code will often have repeating patterns of the type "every fifth letter is an O." If that occurs often in a section of code, then a model that considers the 5th most recent byte will be very good at compressing that

data. Of course, computing these many models with all of their different possible historical contexts is very expensive, meaning it takes a lot of time and computer memory. But it doesn't matter for Crinkler, because it is working on very small amounts of code.

Figure 4: Probability model

data	A	B	C	D	E	F	G	H	*
0 model	0	0	0	0	0	0	0	0	*
128 model	0	0	0	0	0	0	0	1	*
129 model	1	0	0	0	0	0	0	1	*
153 model	1	0	0	1	1	0	0	1	*
255 model	1	1	1	1	1	1	1	1	*

Combining these models ends up being quite easy. Given a set of counters and weights (for a specific context), the formula for the probability that one occurs after that context is shown in this equation:

$$p(1) = \frac{\sum \frac{w * n_1}{n_0 + n_1}}{\sum w}$$

In this equation, n_1 is the one counter for that context (for each model), n_0 is the zero counter (for each model), and w is that model's weight. There are several different ways to determine the weight associated with each model. The algorithm could calculate a weight as the algorithm goes along, which involves neural networks and other interesting computer science features, but Crinkler does not do this because experiments have shown that the decompressor (the part of the code that is attached to the final executable) is made around 100 bytes longer using this method, and its gains in compressibility do not offset that loss. Instead, Crinkler makes rough estimates by setting the weight to the number of bits that are on in the bit mask (the number of ones in a model in Figure 4). Crinkler gives the user an option to use slow compression, which then optimizes these weights a little further.

2.2.3 Choosing Which Models to Combine

Now, Crinkler needs to determine which of these models is the best for each section of code. This procedure—choosing which models fit which sections—is largely why Crinkler acts as a linker; as previously stated, different types of data are split up so that similar data can be grouped together, and models for these data can be stronger. To determine which models are the best to use, the algorithm could, as previously stated, calculate the weights as it goes along and then select the models with the best weights. However, because it doesn't do that, it solves it in a way that is much simpler, if not as ideal. It loops through the models, and for each model, checks whether or not compression improves. After that, it loops through the existing models in the set and checks whether removing any model improves compression. This might be a slower or clunkier solution, but because Crinkler is working with such small files, it doesn't really make a big difference.

Unfortunately, working with so many contexts and models and counters, while great in the theoretical world, is extremely difficult with real data, on real computers. It means that when compressing and decompressing, all of these counters take up a lot of computer memory, which can make programs extremely

slow or completely unusable. To get around this, Crinkler pulls off one last little trick: it uses a hash table. A hash table basically just associates a set of values with a set of keys, so if you have a key you can look into the table and find a value. This improves decompression because Crinkler can turn a distinct historical context into a key by “hashing” it, turning the context into a key, and looking it up in the table, where it stores the context’s probabilities in the form $\frac{n_0}{n_1}$, referencing the two counters from earlier. It can then define the size of the table and keep it from exploding computer memory. Of course, this begs the question: how can a hash table store the same data in a more limited space? The answer is that it doesn’t. Certain contexts can map to the same key even if they are very different, and the chance that this happens goes up as the size of the table goes down. However, Crinkler doesn’t do anything about this problem, choosing to ignore it to keep the size of the decompressing code small. This can throw off the probability calculations for models, making the interval at the end of the arithmetic coding algorithm unnecessarily small, but it doesn’t increase the size by as much as if code to handle collisions were put into the decompressor.

3 Conclusion

Thus, Crinkler compresses intros and demos drastically. It acts as a linker which groups similar data together and optimizes code that Windows, the operating system, needs to run the program; it uses an arithmetic coding algorithm to compress the data; it supplies that coder with models that it calculates and then combines with weights, selecting the best models, and it uses a hash table when calculating probabilities to make sure that the computer can keep track of everything. By artfully combining some of the best lossless compression algorithms with unique software engineering techniques, Crinkler has become one of the most popular demotools in the Demoscene, an invaluable tool used by many to continue to push their technology and skills past their limits.

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The Mathematics Behind the Rubik's Cube

Lamisa Aziz

1 Introduction

Have you ever played with a Rubik's Cube or seen your friends try to solve one? A Rubik's Cube is a 3x3x3 cube with six colors (red, blue, yellow, green, white, and orange). A pivot mechanism inside the cube allows you to twist each face so that the colors can be switched around. You try to solve the cube by getting the colors back together on the same corresponding side. Everyone who uses one usually tries to solve it as fast as possible, yet some fail while others can even do it in under 30 seconds! With all the different colors, they are not only eye-catching cubes of fun but also a research interest. Many want to learn the science and math behind this famous puzzle, but little do they know it could be so intriguing.

2 History

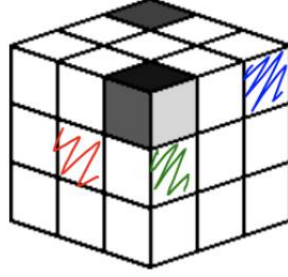
"We turn the cube and it twists us," said Erno Rubik, engineer of this unique design, as he explained the purpose of his creation. Rubik was a socialist bureaucrat who lived in Budapest, Hungary. He built the simple toy in his mother's apartment, not knowing that this invention would have an incredible impact on the future. He was first inspired in spring of 1974 by the popular 15 Puzzle. Invented in the late 1870s, this puzzle consists of 15 consecutively numbered, flat squares that can be slid around inside a square frame. A few years later, Sam Loyd, a recreational mathematician inspired by Erno Rubik, created a two-dimensional Magic Cube. It was later renamed to honor Erno Rubik to the Rubik's Cube. Since then, there have been many different cube variations made, but the one we will be discussing is called the standard 3x3x3.

3 Notation

Behind the 54 colorful little squares is a world of math that determine their possible arrangements, along with the countless ways to solve it. The basic notation for the six faces of the Rubik's Cube is F (front side), B (back), R (right), L (left), U (up), and D (down). Face rotations are represented by F for 90 degrees clockwise, F' for 90 degrees counterclockwise, and F^2 for a 180-degree rotation. To refer to individual 1x1 cubes within the Rubik's Cube, we use one letter to denote the center cube, two letters for an edge cube, and three letters for the corner cube. As shown in the following graphic, we refer to the red cube as at R (right), the green as at DL (down side, left), and the blue cube as at DUR (down side, up, right). The first letter always represents the side (face) of the Rubik's Cube where it is located.

4 Counting the Possibilities

When you try to solve a Rubik's cube, your way will probably be different from the way someone else tries to solve it. Why is that? It is because there are billions of ways to solve it due to the countless possibilities of moves. First, there are eight corner pieces that can be arranged in $8!$ ($8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$) or



40,320 different ways. Each has three orientations (because of the three colors on every corner piece), so there are 3^8 or 6,561 possibilities for each permutation of a corner piece. Then, you have 12 edge pieces with 2 orientations, resulting in 2^{12} or 4,096 permutations. However, there are some restrictions since in the solution for the Rubik's Cube, only $\frac{1}{3}$ of the permutations of the corner cubes are correct, and only $\frac{1}{2}$ of the permutations of the same-edge cubes are correct so that you are brought closer to the final result. The remaining 12 pieces of the cube (27 total pieces - 6 center pieces on the faces - 1 piece in the center of the cube that isn't visible - 8 corner pieces = 12) can be arranged in $12!$ or 4,790,001,600 ways.

Also, there is a $\frac{1}{2}$ correct cube rearrangement parity which results in the following equation which represents the total number of permutations in the Rubik's Cube:

$$\frac{8! \cdot 3^8 \cdot 12! \cdot 2^{12}}{3 \cdot 2 \cdot 2} = 4.3252 \cdot 10^{19}$$

However, after you make a move in the Rubik's Cube, you may also want to go back if you made a mistake. We also have to count those moves to find the precise number of arrangements. The arrangement involves a maximum and minimum number of steps. It is represented by the following equation, where $n \geq 19$:

$$12 \cdot 11^{n-1} \geq 4.3252 \cdot 10^{19}$$

In addition, we can represent cube permutations as a group of elements which we will call R . A group (G) consists of set of objects and a binary operator ($*$) in which it satisfies these four conditions:

1. The operation $*$ is closed, so for any group elements h and g in G , $h * g$ is also in G .
2. The operation $*$ is associative, so for any elements f , g , and h , $(f * g) * h = f * (g * h)$.
3. There is an identity element $e \in G$ such that $e * g = g * e = g$.
4. Every element in G has an inverse g^{-1} relative to the operation $*$ such that $g * g^{-1} = g^{-1} * g = e$.

As a result, our binary operator for the Rubik's Cube represents all the cube moves, sequences or basically, the rotations of a face of the cube. This operation is closed since any face rotation gives us another permutation of the cube also in R . Rotations are also associative: it does not matter how we group them, as long as the order in which operations are performed is conserved. The inverse of a group element g is usually written as g^{-1} .

5 Basic Theorems

Like most mathematics, the Rubik's Cube has its own set of theorems:

1. The cube always has even parity, or an even number of cubies (small cubes within the large cube) exchanged from the starting position.
2. If the cube starts at the solved state, and one move sequence P is performed successively, then eventually the cube will return to its solved state.

These theorems help mathematicians further understand the Rubik's Cube as a whole, along with the math that describes the way it functions. Some of the common properties used to analyze the Rubik's Cube are associative and commutative properties. For the associative property, the permutations in the row can be grouped together. An example could be $(RY')L = R(Y'L)$. For the commutative property, although commutativity is not a necessary condition of the permutation group, it shows when permutations are similar to each other, as in $FB = BF$. In addition, there is a neutral element which means there is a permutation that doesn't rearrange the set, like RR' , and an inverse element which means every permutation has an inverse permutation, like $R - R'$.

6 Research

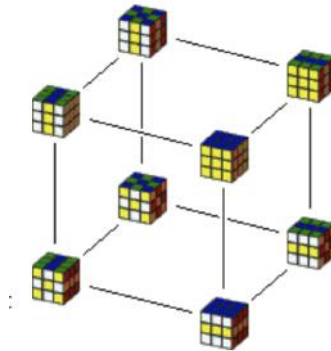
There are many researchers who examined and tested the Rubik's Cube to find the science/math behind it. Erik Demaine, a professor of computer science and engineering at MIT, explained, "The standard way to solve a Rubik's Cube is to find a square that's out of position and move it into the right place while leaving the rest of the cube as little changed as possible. That approach will indeed yield a worst-case solution that's proportional to N^2 ." He and his colleagues had come to the conclusion that there exists a single sequence of twists that could move multiple squares into their proper places, cutting down the total number of moves. The only challenge that really stood in their way was how to describe those circumstances in mathematics, rather than words.

7 Lagrange's Theorem and Cayley Graph

However, an essential theorem used for the Rubik's Cube is Lagrange's Theorem, which states that if H is a subgroup of a finite group G , then the order of H divides the order of $G(|G|)$. This theorem was first developed by Joseph-Louis Lagrange in 1771. Some corollaries that follow from this theorem are:

1. Let G be a group and $g \in G$. Then $|g|$ divides $|G|$.
2. Let G be a group of prime order. Then G has no subgroups and hence is cyclic.

On the other hand, since a picture is worth a thousand words, a great way to gain insight into the structure of groups and subgroups of the Rubik's Cube is to draw a Cayley Graph, as shown below. The graph is unique and contains many properties. When graphing group G , each $g \in G$ is a vertex, each group generator $s \in S$ is assigned a color c_s , and lastly, for any $g \in G$, $s \in S$ elements corresponding to g and gs are joined by a directed edge of color c_s . Drawing a Cayley Graph for all real numbers would be infeasible since it involves about 43 trillion vertices. Instead, we should graph small subgroups of R .



8 Conclusion

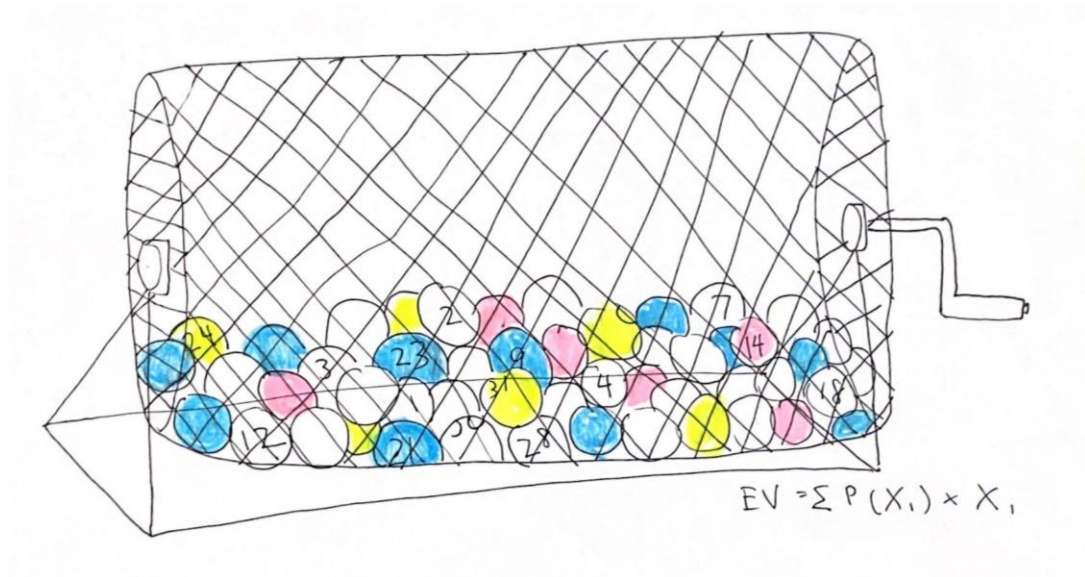
In conclusion, analysis of the Rubik's Cube shows how math is a part of our everyday lives, not just our classes! Most of us only look at the Cube as a way to pass time or as just another fun activity to do with friends and family. Yet we rarely think about all the intriguing math that underlies it. Much more can be written about how math relates to this fascinating little puzzle. Keep looking for applications of math in all aspects of life!

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You “MIT” Just Win the Lottery!

Xiaoshen Ma



1 Introduction

Those who don't skip Physical Education will know that a recent attendance question asked, “What would you do in life if you knew you could not fail?” Readers of *Math Survey* might be thinking of something nerdy, but there's always the classic, go-to answer of winning the lottery. An old saying calls the lottery a “tax on the stupid,” and many regard it as another form of gambling. It should not be surprising then that the first dated lottery system started with betting on elections in 17th-century Genoa. Yet when there are losers, there must be winners. And with some bugs in the lottery's set-up and some interesting calculations, some may win with chance instead of by chance.

2 A Little History

Genoa chose their two *governatori* not by democracy, not by appointment, and not by nepotism – presumably so. They simply drew the names of two people from a heap of 120 slips of paper with the names of 120 councilors. Personally, it was not long after March 15, 2020 before I started betting donuts on the duration of quarantine, so with the first casinos and gambling houses taking over Italy in the 17th century,

it's reasonable that bored Genoan gamblers saw an opportunity in their electoral system. They put a number to each councilor, and eliminated the need to even show up at pubs on Election Day. Sound familiar?

Adam Smith wrote in *The Wealth of Nations* that “there is not, however, a more certain proposition in mathematics, than that the more tickets you adventure upon, the more likely you are to be a loser. Adventure upon all the tickets in the lottery, and you lose for certain.” Hopefully, most of us are familiar with being told that the more you gamble the more you lose. While this may be good advice for my wallet before I have to take everyone out for a Dunkin’ Donuts buffet, the first part of his argument is not necessarily true. One can argue that buying more tickets for the lottery increases one’s chances of winning. Another can argue that buying more tickets only means spending more money. To put these arguments to numbers, you need something called an *expected value*.

3 A Donut Lottery Example

Let’s say there are 20 different flavored donuts to give away. If I happen to give out my calculus teacher’s mystery favorite flavor, he’d buy me 17 donuts in return (not that he would ever dabble in transactions like this). If I give out all 20 donuts, then I surely would make a whopping profit of negative three donuts. This is what Smith meant by “lose for certain.” In comparison, if I only give out one donut, I’d at least have a $\frac{1}{20}$ chance of winning a profit. If I give out an additional one, I’d have a $\frac{2}{20}$ chance of winning. The breaking point is when I give out 17 donuts. The chance of being rewarded by my teacher or basically an all-for-nothing is $\frac{17}{20}$, and there’s a $\frac{3}{20}$ chance that I lose donuts. Until now, I’ve continuously made myself less likely to lose by giving out more. However, the moment I spend another donut, I’m sure to lose despite my teacher’s favorite flavor. Another thing to consider is that giving out 17 donuts makes me less likely to lose, but the consequences of losing become much more grave than giving out one. Expected value assigns a value to the number of donuts given out with the outcome and the chance of the outcome taken into account. Thus, we calculate it by multiplying each possible outcome by the probability of it happening, and summing up all of those products.

$$EV = \sum P(X_i) \times X_i$$

If we get the right flavor: $\frac{1}{20} \times 17 \text{ donuts} = \frac{17}{20} \text{ donuts}$.

If we don’t: $\frac{19}{20} \times 0 \text{ donuts} = 0 \text{ donuts}$.

Expected value of each donut: $\frac{17}{20} \text{ donuts}$.

By participating, I lowered the value of each donut to that of $\frac{17}{20}$ th of a donut. Maybe I should look for donut flavors that are statistically more appealing to math teachers instead.

Now, most lotteries have way more outcomes other than completely winning or losing, in order to keep people going for the smaller prizes. Most also take cash over donuts. Don’t confuse expected value with unit price either. There is no way anybody would measure out $\frac{17}{20}$ th of a donut just for you. Some other things to take into account are that when prizes go unclaimed, more people enter the prize pool, and the jackpot grows. With this, expected value grows as well. Let’s say that many more people are suddenly interested in betting donuts. My teacher is now willing to give out different prizes for different favorite flavors. There’s a:

- $\frac{1}{3300}$ chance of winning the jackpot of x donuts

- $\frac{1}{2500}$ chance of winning 500 donuts
- $\frac{1}{2000}$ chance of winning 400 donuts
- $\frac{1}{1500}$ chance of winning 250 donuts
- $\frac{1}{1250}$ chance of winning 175 donuts
- $\frac{1}{1000}$ chance of winning 17 donuts like the good old days.

Then the expected value of each donut is now: $\frac{x}{3300} + \frac{500}{2500} + \frac{400}{2000} + \frac{250}{1500} + \frac{175}{1250} + \frac{17}{1000}$ and x changes based on the number of participants.

Now we set this greater than or equal to the one donut you spent:

$$\begin{aligned} \frac{x}{3300} + \frac{500}{2500} + \frac{400}{2000} + \frac{250}{1500} + \frac{175}{1250} + \frac{17}{1000} &\geq 1 \\ \frac{x}{3300} + 0.723667 &\geq 1 \\ \frac{x}{3300} &\geq 0.276333 \\ x &\geq 911.9 \end{aligned}$$

And with some simple algebra, you can find the optimal time to invest in an underground donut. In this example, the expected return value is positive once the jackpot exceeds 911 donuts.

4 The Cash WinFall Example

At this point those here to get information on a good college might be wondering if the title is pure clickbait, but in 2005 an MIT student named James Harvey also realized he could do some quick calculations with Massachusetts' Cash WinFall. The rules of the game were:

- Players have six numbers from 1-46.
- You can win by matching with at least two winning numbers drawn every two weeks.
- Getting all six numbers wins you the jackpot.
- The jackpot starts at \$500,000 and goes up every time there's no winner until it reaches \$2,300,000.
 - Once the jackpot exceeds the maximum value, it resets to \$500,000 and the rest of the money gets distributed among the smaller prizes.

Starting from the jackpot, there's a $\left(\frac{1}{46}\right) = \frac{1}{9366819}$ chance of winning. On a normal day with a million dollar jackpot, the expected value of the ticket could've been \$79.80. With the ticket priced at \$2, this wasn't the best trade. On a rolldown day, \$600,000 was distributed among the 5 and 3 number prizes, while \$1.4 million got put into the 4 number prize. Doing our usual calculations, you'd get that now the ticket's expected value is \$5.53, yielding a 167.5% return. Knowing this, students at MIT bought thousands of tickets and soon, tens of thousands. Not long after, the project turned into a thriving business named Random Strategies. The lottery would close seven years later, but the business made \$3.5 million off of this game.

Random Strategies wasn't the first to discover this, nor Doctor Zhang Lottery Club, nor retiree Gerald Selbee. Eighteenth-century France tried all sorts of ways to drive the economy. An awkward time passed by when John Law tried to introduce a paper money system (1716-1720). Nevertheless, under Louis XV, minister Cardinal Fleury stabilized France's monetary system by implementing a conversion between silver, gold, and coins used across the country. Michel Le Peletier des Forts, a deputy officer, wanted to drive the sale of bonds that funded the government. He attached the right to one ticket with a 500,000-livre prize to each bond sold. Unfortunately, Le Peletier might've skipped a few math classes, and made some *silly mistakes* when designing the lottery. Like Cash WinFall, it had rolldown days. Like Cash WinFall, tickets had a great positive return value on these rolldown days. Like Cash WinFall, buying enough tickets would give you a bigger chance of winning quite a sum. Mathematician Charles-Marie de La Condamine saw this even a couple hundred years ago and gathered his buddies. Among them was writer and public activist Voltaire, who actually gained most of his wealth from investing in this lottery system instead of his writings.

It is legally required for meddlers in the lotto to be at least 18 years old in New York, so before you get excited, know that not all lotteries have obvious loopholes, and that I do not endorse getting fake IDs. However, if in a couple years you do find potential in a certain -Fall or -Ball or -Millions, do ring me up.

5 Problems

1) The spinning donut of fortune has five sections. Landing on the blue section gives you \$1000, yellow gives you \$500, white gives you \$100, red gives you \$50, and green gives you another spin. $\frac{1}{50}$ of the donut is blue, $\frac{1}{25}$ of it is yellow, $\frac{1}{20}$ is white, $\frac{1}{10}$ is red, and the rest is green. What is the expected value of each spin?

2) We play a game. The pot starts at \$0. On every turn, you flip a fair coin. If you flip heads, I add \$100 to the pot. If you flip tails, I take all of the money out of the pot, and you are assessed a "strike." You can stop the game before any flip and collect the contents of the pot, but if you get three strikes, the game is over, and you win nothing. Find, with proof, the expected value of your winnings if you follow an optimal strategy. (USAMTS Year 17, Round 4, Problem 3)

3) Bobby Boo is giving out 100 different donuts. There are 95 different non-chocolate flavored donuts and five chocolate donuts. The donuts all look about the same, and Bobby Boo wants to keep one chocolate donut for himself, so he will stop giving out donuts once he gives out the fourth donut. What is the expected average number of donuts Bobby Boo will give out?

4) Blubby Blo put an ant on a 5 x 5 square, and it took four steps (each step crossing to a new square) in a random direction (except back to the square it came from). If the ant ends up in a corner, Blubby Blo wins 10 donuts. If it ends up on an edge, he wins five donuts. Otherwise, he gets nothing. What is the expected number of donuts Blubby Blo will win?

5) Crabby Cat has four blue socks, four white socks, four pink socks, and four leopard-print socks. It's the morning, and Crabby Cat is too lazy to look inside her drawer, so she'll just keep pulling out socks until she has two pairs of matching socks. What is the expected average number of socks Crabby Cat will pull out?

6) Dooby Doo is taking cookies out of the oven without replacement (because he's eating them). There are five cookies in total, and they are numbered from 1 to 5. He'll stop taking cookies out once he takes out a number greater than the one he just ate. What is the expected number of cookies Dooby Doo eats?

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A* Algorithm: Finding an Optimal Path from Point A to Point B

Jason Jiang

1 Introduction

Humans have had to figure out the best route from Point A to Point B throughout history. Having been born and raised in Brooklyn, I've never been scared by a lost tourist approaching me on the streets. It's easy for us humans to retrace a previous path and guide others in the right direction.

Fast forwarding to the 21st century, the computers all around us can replicate this optimization. It's found in our video games' characters who move to our cursors, our self-driving cars, and even our vacuums. The best-known example is probably Google Maps, which gives extremely fast and accurate travel information for any route. There are many methods to solve this problem, but I will be diving into just one of the ways a computer can find an optimal path, the A* Algorithm.

2 History and Uses

A* Algorithm was created by the efforts of Peter Hart, Nils Nilsson, and Bertram Raphael back in 1968. It was first used to build faster pathfinding AI for a robot called Shakey the Robot. The implications of their discovery were not just limited to robotics. Pathfinding algorithms are useful for sequence games like Othello. They are also used in path movement for video games that move the player to the cursor. They help with computer traffic and graph traversal. Their applications are incredibly varied! Anywhere that you have different paths and want to know how to get from Point A to Point B, the A* Algorithm is your best friend!

3 How it Works

A* Algorithm looks at undiscovered places to go, using a heuristic that estimates the total cost of going from the origin through the node to the goal, evaluating the most promising nodes first.

1. The A* Algorithm starts at the origin, looks at all the undiscovered points that it can reach in one move, and then adds them to a queue with the parent property as the origin and keeps track of the cost to get there.

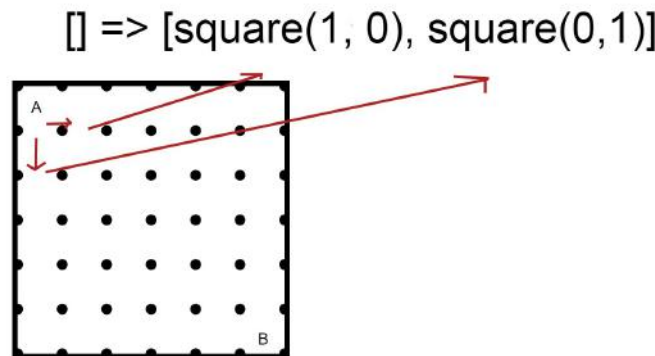


Figure 1: Step 1: The squares next to the starting point (A) are added into the queue

2. This is when the “secret sauce” of the A* Algorithm gets to work. The queue is sorted by adding together the cost to get to the node and the estimated distance from the destination. Together this forms an estimate of the distance that the most optimal path would take going through that node:

$$f(n) = g(n) + h(n)$$

f = total estimated cost to go through node

g = cost to get to the node

h = estimated distance to get to the destination

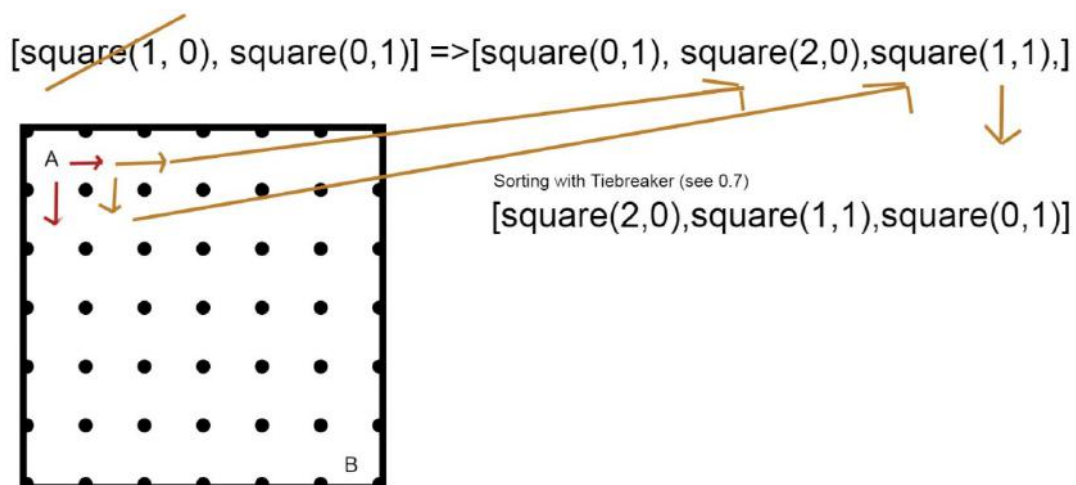


Figure 2: Step 2: From the origin, A* looks at square(1,0) and finds the nearby squares: square(2,0), square(1,1). It then appends and sorts them into the list of undiscovered squares.

3. The A* Algorithm then repeats the first two steps, looking at the undiscovered nodes around the first element in the sorted queue instead of the origin. It then continues evaluating these nodes until either the program reaches the destination or a path is not found.

4. At the end point, the program can backtrack to each of the parent nodes until it reaches the origin, constructing the solution path and calculating its distance. Every node contains a back pointer which stores the previous node in the path.

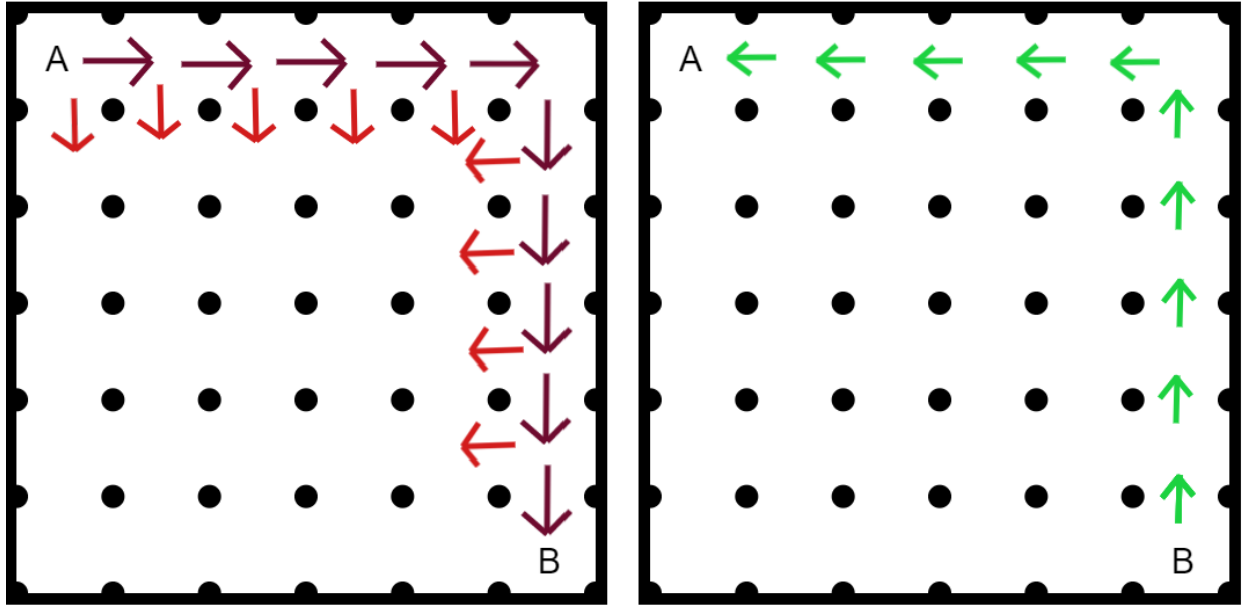


Figure 3: Steps 3 and 4: A* searches through the grid, prioritizing nodes that move closer to the destination. After reaching the end point, it retraces its steps to find the optimal path.

4 (When) does A* give us the correct shortest path?

In a specific implementation of A*, the answer to this question is: always!! To prove it, we rely on these two assumptions:

1. We have an admissible heuristic!! In the sorting process, the heuristic function estimates the cost of reaching the goal through a node. An admissible heuristic ensures that the estimation of the total cost will never be above the actual total cost, which is distance of the path drawn from the starting point, through the node, to the destination. (See the green line in visualization).
2. The heuristic value of the node is the same anywhere along its estimated optimal path. So, when A* Algorithm sorts the queue and picks the next best node, we can think of that as the A* Algorithm “trying out” this optimistic estimated path.

Thus, because A* Algorithm is always “trying out” the optimistic estimated path every time, it sorts the queue and picks the best node. We know that when A* Algorithm reaches the end point, it will have chosen the best path.

5 Manhattan Distance and Tie Breakers

A*'s performance and success is determined by the strength and accuracy of your heuristic, so having a robust one is important. You can choose from a variety of heuristics, but in my visualization, I used the Manhattan Distance, a well known heuristic that is amazing at working with grids. The formula is very simple:

$$\text{abs}(\text{goal}.x - \text{node}.x) + \text{abs}(\text{goal}.y - \text{node}.y)$$

The reasoning behind it is that no matter what, the path to reach the goal has at least a displacement of $\text{abs}(\text{goal}.x - \text{node}.x)$ and at least a displacement of $\text{abs}(\text{goal}.y - \text{node}.y)$.

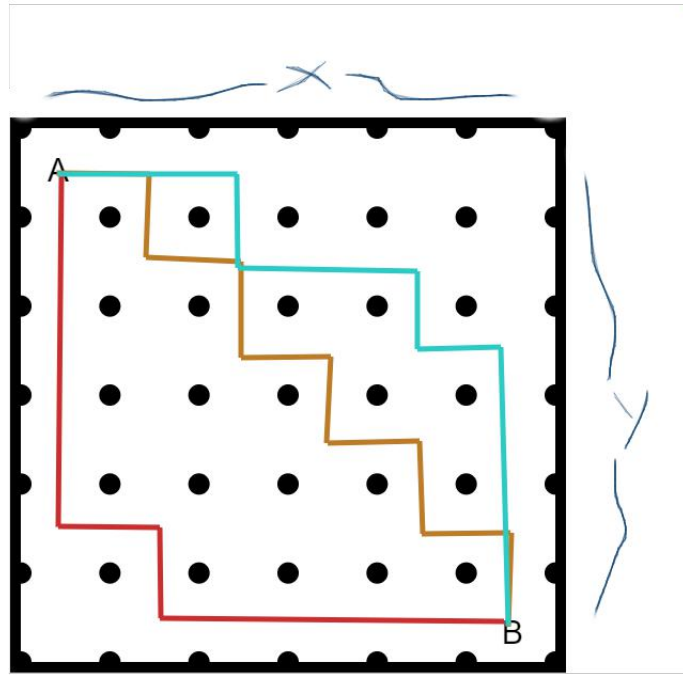


Figure 4: All these paths have the same Manhattan Distance

However, as we see in Figure 4, there can be many paths that all have the same Manhattan Distance. If they all have the same Manhattan Distance, this slows down A*, since it attempts to go down all these paths. We need some way to tell A* to prioritize just one of these equivalent paths and stick with it.

That's when tie breakers come in. To implement a tie breaker, we will create a conditional. If there is a tie among the estimated cost between nodes, A* will favor the nodes closest to the goal. Therefore, A* will not waste time exploring different but equally optimal paths.

6 Conclusion

A* is a simple but powerful pathfinding algorithm that is used in many kinds of applications. I hope this short introduction inspires you to utilize A* for your next project. Next time you flick on your phone to turn on Google Maps, pat yourself on the back because you can now appreciate some of the optimizing behind calculating your route.

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Fermat's Little Theorem and Cryptography

John Gupta-She



1 Introduction

Computer science is math. Everything about computer science is math. Computer science is just another form of math, one that lets us handle bigger numbers that we'd otherwise be scared of. Of course, encryption is no different than the rest of computer science: it uses a certain concept in math, number theory. More specifically, the concept of encryption is the concept of Fermat's Little Theorem (FLT).

2 Fermat's Little Theorem

Of course, to understand how FLT relates to encryption, we have to explore it in detail first. FLT is a basic, but essential, number theory concept that uses prime numbers' unique condition of being relatively prime to all other natural numbers. The statement goes like this:

If p is a prime, then

$$a^{p-1} \equiv 1 \pmod{p}$$

This is a nontrivial fact, so it must be proved. The proof is quite clever, and as mentioned before, leverages the fact that primes are relatively prime to all other natural numbers.

2.1 Proof

Consider the sets $\{1, 2, 3, \dots, p-1\}$ and $\{a, 2a, 3a, \dots, (p-1)a\}$. I claim that these sets are equivalent mod p as long as a is relatively prime to p . To prove that, we can consider the second set. We know the first set is all the numbers from 1 through $p-1$, so we just need to prove that the second set is equal to that, mod p . With mod p , we already know that if there are no repeats, then there are only p different possibilities for elements of this set: 0 through $p-1$. We also cannot get 0, because that implies one of ja is divisible by p , where j is a random number from 1 through $p-1$. This means that if all the elements of $\{a, 2a, 3a, \dots, (p-1)a\}$ are different, we will have $p-1$ distinct elements to fill $p-1$ spaces (namely 1 through $p-1$), which is exactly what we want. Now let us prove that all terms of $\{a, 2a, 3a, \dots, (p-1)a\}$ are distinct. Let us assume that two elements are congruent mod p . Let us call them ia and ja . That means that

$$ia \equiv ja \pmod{p}$$

$$ia - ja \equiv 0 \pmod{p}$$

Since a is relatively prime to p , we can divide by a on both sides:

$$i - j \equiv 0 \pmod{p}$$

Since i and j are each between 1 and $p-1$, we know that their difference is between $-(p-2)$ and $p-2$. The only number in this range that is $0 \pmod{p}$ is 0, so their difference must be 0. This means that $i - j = 0$, so $i = j$. This is a contradiction, since we assumed that these were two different elements of $\{a, 2a, 3a, \dots, (p-1)a\}$, but these are the same element. Thus, our original assumption that there are two elements that are the same mod p is incorrect, so they must all be different mod p . Remember that this proves that the set $\{a, 2a, 3a, \dots, (p-1)a\}$ is really the same as the set $\{1, 2, 3, \dots, p-1\}$. We know that two equivalent sets mod p must have the same product mod p . This gives

$$(p-1)!a^{p-1} \equiv (p-1)! \pmod{p}$$

Since all of the numbers from 1 to $p-1$ are relatively prime to p , we can divide by $(p-1)!$ on both sides, resulting in:

$$a^{p-1} \equiv 1 \pmod{p}$$

Thus, we have arrived at the theorem statement, so the proof is completed.

2.2 Euler's Generalization

There is a generalized version of FLT that does not rely on the mod being prime, but rather that the base (a) and the mod (let's call it k) must be relatively prime. It is a proof that is rather similar to the proof I presented here, so I will leave it as an exercise. Just remember the main points of the previous proof: two equivalent sets mod k that are also a multiple of a away from each other. In the end, instead of the power being a simple expression like $p-1$, it becomes the output of a function on k , that we call Euler's totient function, denoted as $\phi(k)$.

3 Connections to Encryption

It's difficult to make a clear connection to encryption until you really apply this concept. Let's explore an encryption scenario that is strong because of its usage of FLT.

Say we want to send a message that is an integer mod N . Choose an integer a from between 1 and N that is relatively prime to $\phi(N)$. The public key is made up of N and a where a is called an encryption key.

At this point, anyone can use this public key to encrypt messages to the creator of the public key, because the creator will have a private key to decode messages. To encode the message, take a number you wanted to encrypt, let's call it M , raise it to the a th power, and take it mod N . This would make a new number mod N that seemingly has no relation to M , whatsoever. The new number would be

$$A(M) \equiv M^a \pmod{N}$$

where A is some sort of function.

To decode this, we need to determine what this private key will look like. The private key, b , should satisfy the condition

$$a \cdot b \equiv 1 \pmod{\phi(N)}$$

This implies that

$$a \cdot b = \phi(N) \cdot k + 1$$

Now to decode $F(M)$ back into M , mod N , we raise $F(M)$ to the b power, and take it mod N again. Let the function $B(x)$ equal taking X to the b power mod N . We get

$$B(A(M)) \equiv (M^a)^b \pmod{N}$$

$$(M^a)^b \equiv M^{a \cdot b} \pmod{N}$$

From our previous equation we can replace $a \cdot b$ with $\phi(N) \cdot k + 1$.

$$M^{a \cdot b} \equiv M^{\phi(N) \cdot k + 1} \pmod{N}$$

$$M^{\phi(N) \cdot k + 1} \equiv (M^{\phi(N)})^k \cdot M \pmod{N}$$

We know that $M^{\phi(N)}$ is equivalent to 1 mod N , so

$$(M^{\phi(N)})^k \cdot M \equiv (1)^k \cdot M \equiv M \pmod{N}$$

This shows that $B(A(M)) \equiv M \pmod{N}$, so we do have a valid private key. This all works because of Euler's totient function and the fact that there is an inverse to a . However, finding an inverse to a is not quite as simple if you don't already know the prime factorization of N , like the creator of the public key does.

3.1 Difficulty in Factoring

For someone other than the creator of the public key, they would have to invert $a, \text{mod}(\phi(N))$ to find the private key. Remember, $\phi(N)$ is dependent on the prime factorization of N . For a very large N , this prime factorization can be nearly impossible to find. Think about the amount of calculation needed to prime factor a number. First, you need to check if it is divisible by every prime up until the number's square root. This takes days upon days on its own for a large N . What makes it even worse is that you have to do this process over and over until you know you have it completely prime factored. This takes an absurdly long amount of time and is therefore not a viable way to crack the code.

Without this prime factorization of N , we cannot find $\phi(N)$ or find the inverse of $a(\text{mod}\phi(N))$. In the end, we cannot decrypt the code this way.

3.2 Some of the Largest Primes

Below are some of the largest primes we know currently, so you can see that it will take days and days, even for the world's strongest supercomputer. From largest to smallest, these are the five largest primes we know of:

1. $2^{82589933} - 1$
2. $2^{77232917} - 1$
3. $2^{74207281} - 1$
4. $2^{57885161} - 1$
5. $2^{43112609} - 1$

An interesting pattern we have is that they are all one less than some power of 2. This actually is not a coincidence. These kinds of primes ($2^n - 1$) are called Mersenne primes. The reason we have so many inconceivably large Mersenne primes is because there is actually an organization dedicated to finding the largest Mersenne primes. The organization's name is GIMPS, which stands for Great Internet Mersenne Prime Search. Anyone can download their free software and start looking for Mersenne primes on their own. The part I find the most funny is that the URL for their website is simply www.mersenne.org. Anyway, this organization is responsible for finding much more than just the largest five primes. They are responsible for the largest eight primes, but responsible for the last fifteen record holders for largest prime at their times.

The numbers themselves aren't too interesting, but to really grasp how large these numbers are, you have to look at the number of digits they have. Here we have the number of digits of the first 10 largest primes:

1. 24,862,048
2. 23,249,425
3. 22,338,618
4. 17,425,170
5. 12,978,189
6. 12,837,064

7. 11,185,272
8. 9,808,358
9. 9,383,761
10. 9,152,052

Going back to the original point of this tangent about large primes, even if we choose only two of these extremely large primes, we are already in the millions of digits. That means to decrypt the message, a person would have to check every single prime up until the square root of N , which should also have millions of digits.

4 Final Thoughts

All in all, FLT is an important outcome of number theory, not only for further advances in mathematics, but also further advances in cryptography. The use of this theorem makes it possible to encrypt messages with the use of primes, by using the fact that factoring is not as simple as multiplying (as far as we know). When primes become monstrously large, prime factoring larger numbers is not an easy task, which is the reason this form of encryption is so powerful.

And of course, what kind of article would this be without leaving you with some questions to think about? And here they are: What are some other ways you could leverage these concepts to encrypt information? Is it possible to use a different procedure while still using the core idea of not being able to easily factor large numbers? What would such a procedure look like?

I encourage you all to look more into this field of cryptography. Understanding how your privacy is protected is important, so you know your critical information is actually safe.

5 Problems

1. What are the last two digits of $7^{39} + 9^{41} + 11^{43}$?
2. Define $f(x)$ as x^{x^x} . What is $f(18) + f(19) + f(20) \pmod{91}$?
3. Find $111\dots111$ (97 ones) $\pmod{97}$.
4. Find the smallest positive integer n such that n^{93} and 93^n leave the same remainder when divided by 100. (Art of Problem Solving)
5. Find all positive integers n between 0 and 1000 such that $n^{100} \equiv 1 \pmod{19}$.
6. One of Euler's conjectures was disproved in the 1960s by three American mathematicians when they showed there was a positive integer such that $133^5 + 110^5 + 84^5 + 27^5 = n^5$. Find the value of n . (AIME 1989 #9)

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Dabbling with Desmos: When Polar and Cartesian Graphs Collide

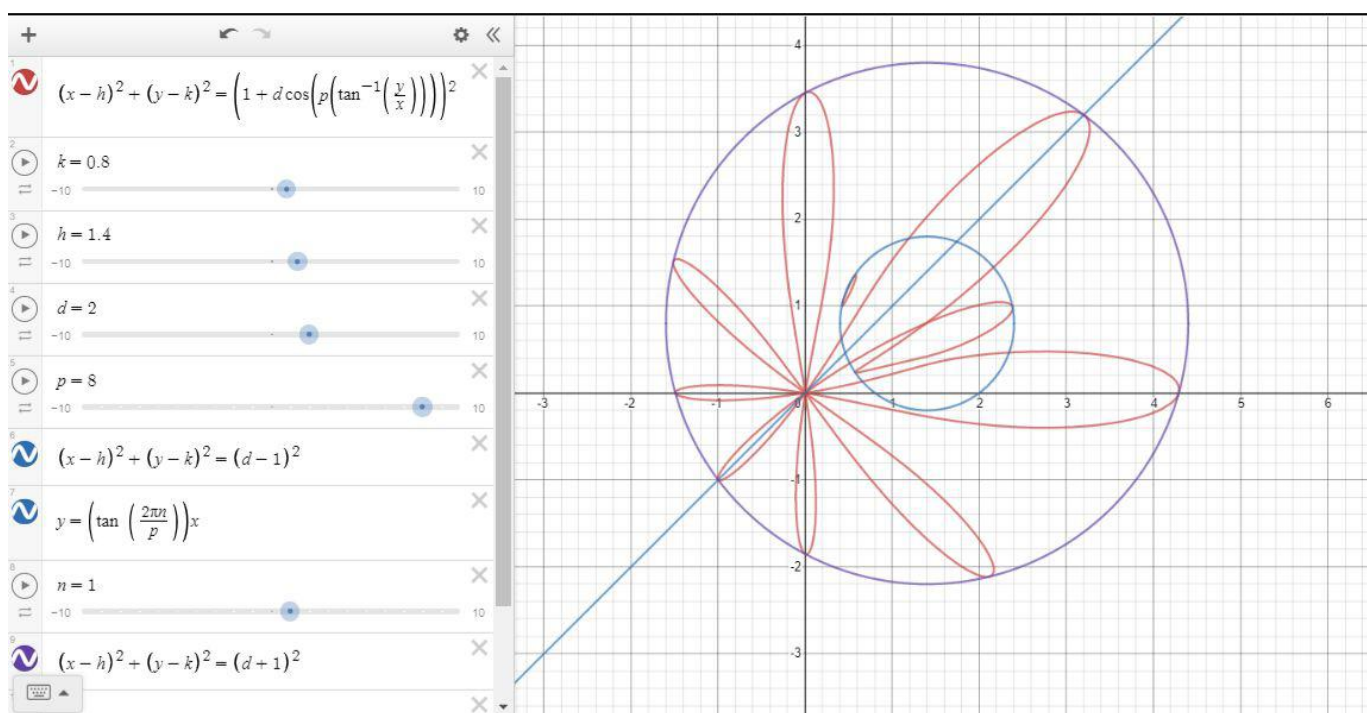
Nora Miller

Some of the most beautiful graphs in math are undoubtedly polar equations. I discovered this inadvertently, as I was trying to create a flower-like image on Desmos, the online graphing calculator. Looking for a method to wrap the periodicity of a sine graph radially around a circle, I first tried replacing both the x and the y in the equation of a circle with sine expressions. Though understandably this did not work, that was the point at which I realized I needed to use polar coordinates.

Polar coordinates, in contrast to Cartesian coordinates, are determined by a point's distance from the origin, and angle relative to the positive x -axis. As a result of their radial nature, they lend themselves to rotation. However, since the Cartesian system lends itself better to translation, orthogonal reflection, and dilation, there is benefit in combining the two. Thus, my first thought after creating a flower in polar coordinates was to convert it to Cartesian coordinates. This process led me to the creation of two intriguing graphs, as I experimented with a variety of mathematical transformations.

1 The Shifting Flower

In order to convert polar coordinates to Cartesian coordinates, I replaced r^2 with $x^2 + y^2$, and I wrote θ as $\tan^{-1} \frac{y}{x}$, to get the equation $x^2 + y^2 = (1 + d(\cos(p \tan^{-1} \frac{y}{x})))^2$. From here, I started replacing $x^2 + y^2$ with $(x - h)^2 + (y - k)^2$, in an attempt to translate the flower. However, before I could replace $\tan^{-1} \frac{y}{x}$ with $\tan^{-1} \frac{y-k}{x-h}$, I realized that the graph I had created looked nothing like a flower.

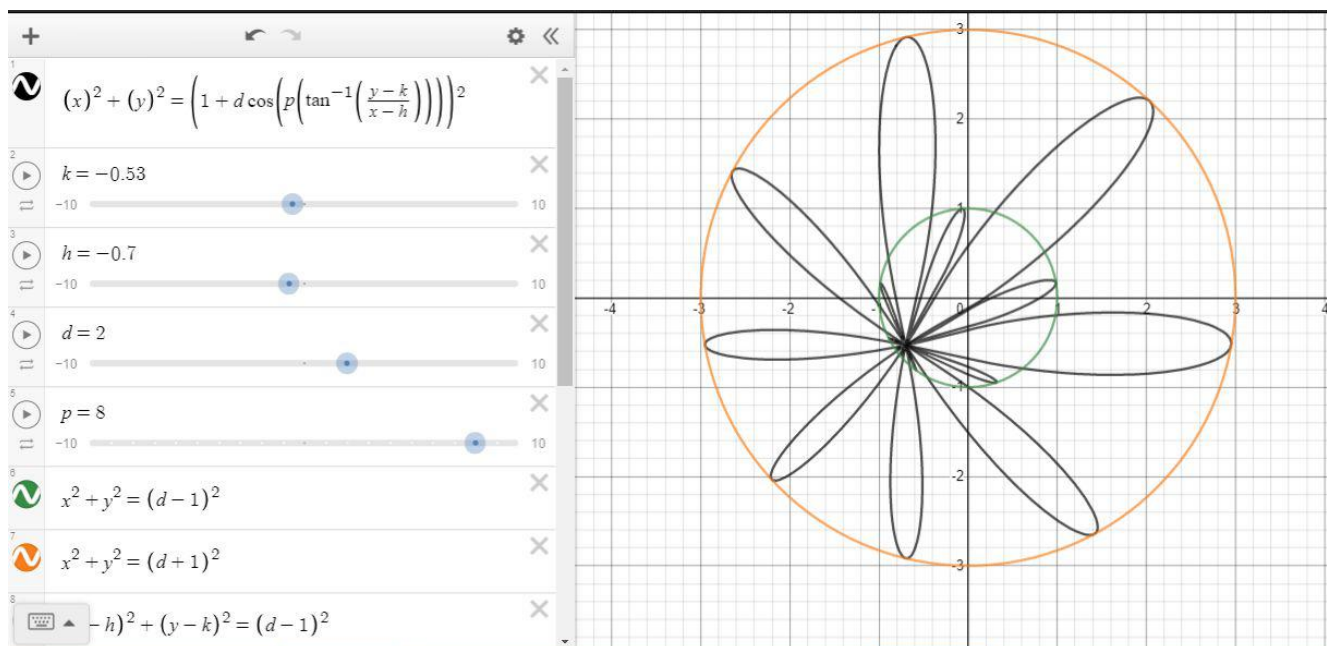


The first thing I noticed about this graph was its uneven petals; provided that the graph was sufficiently close to the origin, the shape had petals. Interestingly, regardless of the value of k , h or d , the petals seemed to point in the same direction, as long as p and a were constant, and they always extended from the origin. This can be thought of in terms of the two main elements of the graph. Looking at the left side of the equation, the coordinate plane can be divided radially into p sections. In each section, $d(\cos(p \tan^{-1} \frac{y}{x}))$ should run through the same set of values, ranging from $-d$ to d depending on the angle. Without translations, this

The image shows a Desmos graphing calculator interface with the following components:

- Left Sidebar (Equations and Sliders):**
 - Equation 1: $(x-h)^2 + (y-k)^2 = \left(1 + d \cos\left(p\left(\tan^{-1}\left(\frac{y}{x}\right)\right)\right)\right)^2$
 - Equation 2: $k = 0.8$ (Slider from -10 to 10)
 - Equation 3: $h = 0.2$ (Slider from -10 to 10)
 - Equation 4: $d = 0.1$ (Slider from -10 to 10)
 - Equation 5: $p = 10$ (Slider from -10 to 10)
 - Equation 6: $(x-h)^2 + (y-k)^2 = (d-1)^2$
 - Equation 7: $y = \left(\tan\left(\frac{2\pi x}{p}\right)\right)x$
 - Equation 8: $n = 1$ (Slider from -10 to 10)
 - Equation 9: $(x-h)^2 + (y-k)^2 = (d+1)^2$
- Main Graph Area:**
 - A purple circle centered at $(h, k) = (0.2, 0.8)$ with radius $d = 0.1$.
 - A blue circle centered at the origin $(0, 0)$ with radius $p = 10$.
 - A red limaçon curve defined by the equation $(x-h)^2 + (y-k)^2 = (d-1)^2$.
 - A blue line $y = \tan\left(\frac{2\pi x}{p}\right)x$.

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Another feature of this graph is its two sets of petals. As d nears 1 then exceeds it, $(x-h)^2 + (y-k)^2 = (1-d)^2$ becomes a point, and then begins to expand again. This is because when $d > 1$, it is possible for $(1 - d(\cos(p \tan^{-1} \frac{y}{x})))$ to become a negative value, with a minimum of $1 - d$. Graphically, when a point is negated, it is inverted — wherever a point might have been relative to (h, k) along a given line, it is now in the opposite direction from along the same line. As a result, the points in a graph with $d < 1$ that were between petals and tangent to $(x-h)^2 + (y-k)^2 = (1-d)^2$ now become the tips of another set of petals extending from (h, k) with a maximum radius of $d - 1$.

A final feature of this graph is the symmetrical spacing of petals, over the y -axis. While this is not an intended feature of my graph, it took me some time before I realized how to change it. This realization came while working on another graph.

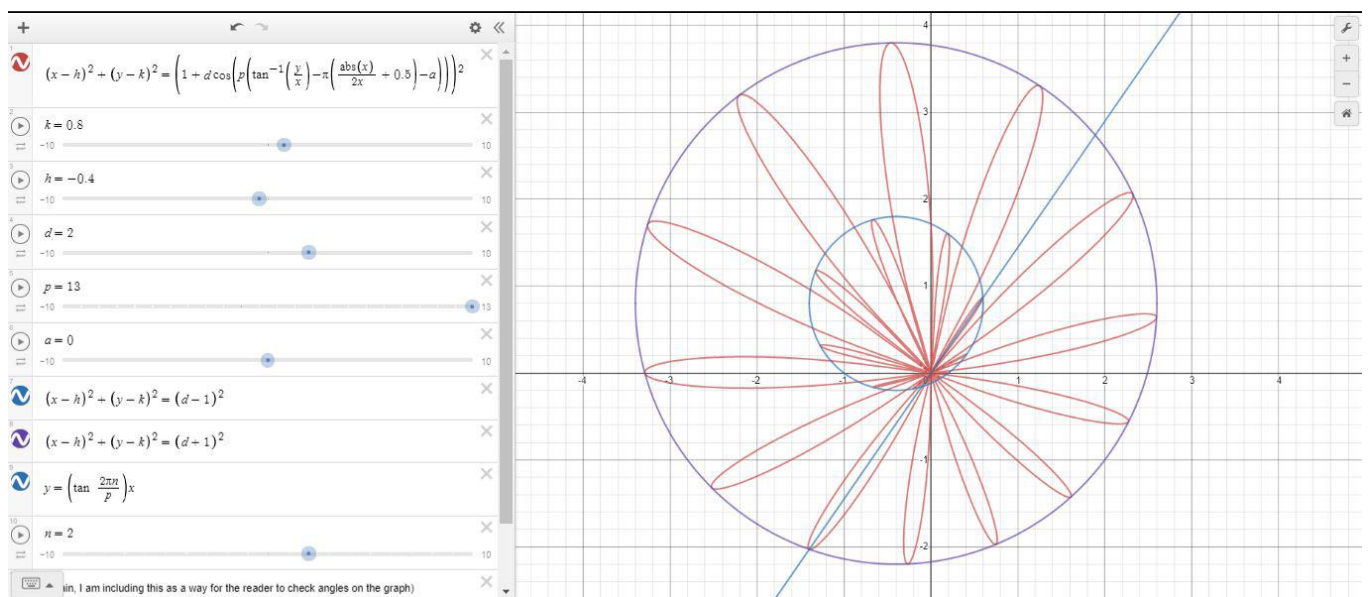
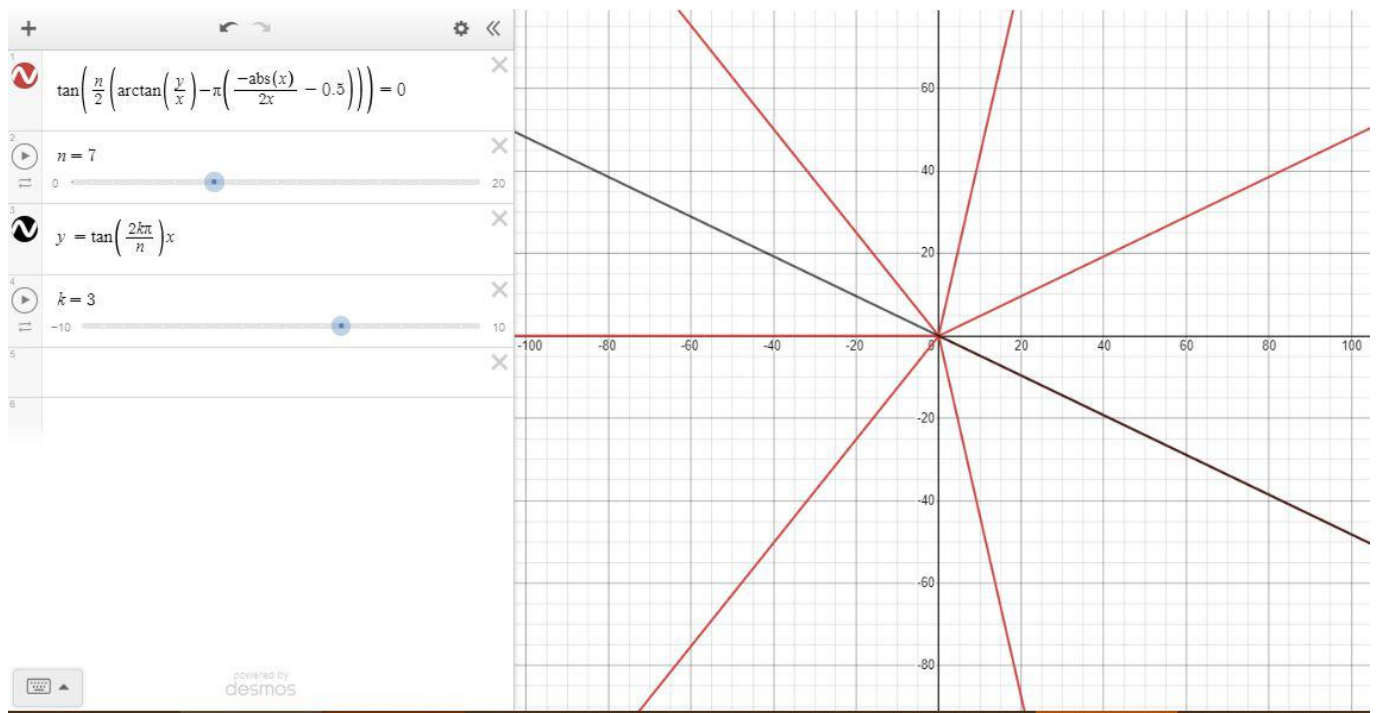
2 Rays, More Efficiently

While working with polar coordinates, I realized that it would be an interesting challenge to divide the plane into equal angles, using multiple evenly-spaced rays. Such a graph would be extremely useful for constructing other radial graphs. I approached this objective using another main advantage of polar coordinates, setting up an equation with angles as solutions. However, this can be difficult, due to the constraints of converting polar to Cartesian coordinates.

My first attempt at evenly-spaced rays extending from the origin involved the polar equation, $\tan(\frac{n}{2}\theta) = 0$. Though Desmos can't graph this, it gives a total of n solutions in the form $\theta = \frac{2\pi}{n}$, with rays on both sides of the y -axis. However, the standard way of converting polar to Cartesian coordinates involves replacing θ with $\tan^{-1} \frac{y}{x}$, which only produces the desired outputs for angles between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$. For angles outside of this range, the inverse tangent function instead gives an angle π apart from the intended angle. As a result, any attempt to convert this graph to Cartesian coordinates results in a graph where quadrants I and IV are accurate, but quadrants II and III are simply a reflection of I and IV over the origin. This is a problem for graphs with an odd number of evenly-spaced rays, but it can be fixed by graphing the rays to the right of the origin separately, and adding π to the output of $\tan^{-1} \frac{y}{x}$ so that negative angles are corrected to positive ones.

However, this addition of π didn't fully resolve the issue. While it is possible to graph every set of rays in two parts, one to the left of the y -axis and one to the right, my intention had been to graph all of my rays together. In other words, I needed a way to rotate the graph by π only in some situations, specifically when $x < 0$. Fortunately for us, the function $\frac{|x|}{x}$ produces two distinct values, -1 and 1 , depending on the sign of x . By dilating this function vertically by -0.5 and translating it up, we get $y = -\frac{|x|}{2x} + 0.5$, which

produces 1 if x is negative and 0 if x is positive. Rotating the value of $\tan^{-1} \frac{y}{x}$ by $\pi(-\frac{|x|}{2x} + 0.5)$ results in a more accurate conversion from polar to Cartesian coordinates, so I substituted $\tan^{-1} \frac{y}{x} - \pi(-\frac{|x|}{2x} + 0.5)$ for $\tan^{-1} \frac{y}{x}$ in my graph of the rays.



In conclusion, using a combination of polar and Cartesian coordinates, I continue to surprise myself with a variety of graphs. While my original goals were to create an image, preferably involving bicycles and/or flowers, for the Desmos Global Math Art Contest, I have come to see the value of creating graphs that are not just visually, but also mathematically, unique.

Diophantine Equations: The Answer Really Is 42

Josephine Lee

1 Introduction: Sum of Cubes Question

Deep Thought's conclusion found in Douglas Adams' *The Hitchhiker's Guide to the Galaxy* - that "the answer to the great question... of life, the universe and everything... is... forty-two" — was paradoxically reflected in completing the final piece of the puzzle of a question mathematicians have been ruminating on since 1954. The question was formulated in Oxford University in 1954 by Professor Louis Joel Mordell: what integer solutions are there to the equation

$$x^3 + y^3 + z^3 = k,$$

where k is less than or equal to 100?

2 History

Answers were most quickly discovered for small cases:

By Fermat's Last Theorem, for $k = 0$, only trivial solutions ($a^3 + (-a)^3 + 0^3 = 0$) exist.

Similarly, for $k = 1$ and $k = 2$, infinite solutions exist:

$$(9b^4)^3 + (3b - 9b^4)^3 + (1 - 9b^3)^3 = 1$$

(discovered by K. Mahler in 1936)

$$(1 + 6c^3)^3 + (1 - 6c^3)^3 + (-6c^2)^3 = 2$$

(discovered by A.S. Verebrusov in 1908)

Over several years, using sophisticated computers, findings were computed for nearly every $k < 100$. We can walk through the algorithm used to find the solutions for $k = 39$ in 1993:

If $k = 3$ then $k \equiv 3 \pmod{9}$, so $x \equiv y \equiv z \equiv 1 \pmod{3}$. If x, y, z are the same sign, $x = y = z = 1$. Otherwise, assume that x, y have the same sign and z have the opposite sign, which makes $|x + y| \geq |z| \geq 1$. We can solve $z^3 \equiv 3 \pmod{3}$ such that $n = x + y$. Since z and n have different signs, $1 \leq |z| \leq |n|$. From the findings in 1989 by D. R. Heath-Brown, we know that $(n, 3) = 1$, and that $n = a^3 + 3b^3 + 9c^3 - 9abc$ such that a, b, c are integers and

$$z \equiv (3c^2 - ab)(b^2 - ac)^{-1} \pmod{n},$$

where $(b^2 - ac, n) = 1$, and z and n have different signs. Then, we find the solutions to $x + y = n$ and $x^3 + y^3 + z^3 = n$ to find unique values for z , yielding:

$$x = \frac{n+d}{2}, y = \frac{n-d}{2}, d = \sqrt{D}, \text{ and } D = \frac{1}{3}[4(\frac{3-z^3}{n} - n^2)]$$

where D is a square root of an integer.

In 1993, researchers used the Cyber 205 vector computer to run a vectorized version of the Euclidean algorithm. Arithmetic operations were able to be vectorized, and the algorithm became capable of solving the equation $w_i x_i \equiv 1 \pmod{n}$, $i = 1, 2, \dots$, at vector speed. It was through this algorithm that the first solution for $k = 39$ was found: (134476, 117367, -159380).

Throughout the decade, the unsolved values of $k \leq 100$ eventually decreased to 33 and 42.

3 Finding the Answers for $k = 42$ and $k = 33$

In 2019, Andrew Brooker from the University of Bristol used a supercomputer to finally find the solution for 33, but the solution for 42 remained elusive. Brooker decided to ask for help from Professor Sutherland from MIT, who used Charity Engine to find the answer for $x^3 + y^3 + z^3 = 42$. Charity Engine utilizes more than 500,000 home PCs with unused computer power, and it still took over a million hours to find the solution. The final successful run took a few weeks.

Employing otherwise unused computational resources makes the Charity Engine network more energy-efficient. The Charity Engine takes advantage of the large portion of the CPU that would otherwise go to waste, making the amount of electricity needed to conduct the computations much less than what would be needed for a supercomputer to do the same work. Brooker and Sutherland also applied several adaptations and optimized the code to make it run more efficiently for a gigantic distributed computation. In contrast, a supercomputer would have to make a long computation run.

The core of the algorithm the Charity Engine uses is as follows:

Suppose that (x, y, z) is a solution to $k = x^3 + y^3 + z^3$ and WLOG that $|z| \leq |y| \leq |x|$. Then:

$$kz^3 = x^3 + y^3 = (x+y)(x^2 - xy + y^2).$$

When $k - x^3 = 0$, $y = -x$, leading to every value of x being a solution.

We can set $d = |x + y| = |x| + y \operatorname{sgn} x$, where

$$\operatorname{sgn}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Thus, d divides $|k - z^3|$:

$$\frac{|k - z^3|}{z} = x^2 + xy + y^2 = |x|(2|x| - d) + (d - |x|)^2 = 3x^2 - 3d|x| + d^2$$

so that the solutions to x and y are

$$\frac{1}{2}\text{sgn}(k - z^3)(d \pm \sqrt{\frac{4|k - z^3| - d^3}{3d}})$$

So, when we run through all the divisors of $|k - z^3|$, we can find all corresponding values of x and y with $\min(|x|, |y|, |z|) \leq B$ in time $O(B^{1+\epsilon})$.

With the solution for $k = 33$, the search bound B was 10^{16} , but for 42, B had to equal 10^{17} , making it much harder for a supercomputer to solve.

4 What Is Still Uncovered in the Sum of Cubes Problem

The Charity Engine found solutions for $k = 42, 165$, and 906:

$$(-80538738812075974)^3 + 80435758145817515^3 + 12602123297335631^3 = 42,$$

$$(-385495523231271884)^3 + 383344975542639445^3 + 98422560467622814^3 = 165, \text{ and}$$

$$(-74924259395610397)^3 + 72054089679353378^3 + 35961979615356503^3 = 906$$

The only remaining unsolved cases up to 1,000 are 114, 390, 579, 627, 633, 732, 921 and 975.

5 Other Diophantine Equations

5.1 The Hardy-Ramanujan Equation

The equation that all taxicab numbers satisfy is

$$w^3 + x^3 = y^3 + z^3$$

where w, x, y, z are all integers.

Here are all the known taxicab numbers:

$$2 = 1^3 + 1^3$$

$$1729 = 1^3 + 12^3 = 9^3 + 10^3$$

$$87,539,319 = 167^3 + 436^3 = 228^3 + 423^3 = 255^3 + 414^3$$

$$6,963,472,309,248 = 2421^3 + 19083^3 = 5436^3 + 18948^3 = 10200^3 + 18072^3 = 13322^3 + 16630^3$$

$$48988659276962496 = 38787^3 + 365757^3 = 107839^3 + 362753^3 = 205292^3 + 342952^3$$

$$24,153,319,581,254,312,065,344 = 582162^3 + 28906206^3 = 3064173^3 + 28894803^3$$

5.2 The Erdős–Straus Conjecture

The Erdős–Straus Conjecture states that for every integer $n \geq 2$, there exists positive integers x, y , and z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

The conjecture has yet to be proved, as of 2019. The most progress we have is from Mordell in 1967, that a counterexample must be one of six infinite arithmetic progressions modulo 840.

6 Applications of Diophantine Equations

Let’s look at a few applications of Diophantine equations:

6.1 Solar System

When investigating the stability of the solar system, the near resonance of the frequencies of planets are associated with denominators.

The dynamical properties of a rotation of angle α on a circle of length L are radically different depending on whether the parameter $\frac{\alpha}{L}$ is rational or not. If $\frac{\alpha}{L}$ is rational, the orbits are all periodic; if not, they are all dense.

Carl L. Siegel, a German mathematician who specialized in analytic number theory, discovered a suitable diophantine inequality on the frequencies to ensure that they are not too close to resonance. His theorem states as follows:

All normalized f with multiplier λ in a subset of full measure of the circle have a guaranteed radius of linearization. The bound is dependent only on the diophantineness order of λ . A diophantine number θ of order $v + 1$ has the property

$$|\theta - \frac{m}{n}| > \frac{Q}{n^{v+1}}$$

where m is the nearest integer to $n\theta$, and $|\lambda^n - 1| = \Omega(\log n)$ as $n \rightarrow \infty$.

6.2 Computational Complexity Theory

Another application of Diophantine equations lies in computational complexity theory. Matiyasevich’s theorem states that a set is recursively enumerable if and only if it is also Diophantine. Thus, the complexity class of recursively enumerable problems are also associated with Diophantine equations, such as the Halting Problem:

Given a computer program and an input, will the program terminate or will it run forever?

The halting set that represents the Halting Problem is

$$K = (i, x) \mid \text{program } i \text{ halts when run on input } x$$

As stated before, the halting set is recursively enumerable; in other words, there exists a computable function that lists all the pairs (i, x) it contains.

Sadly, as proved by Alan Turing in 1936, a general algorithm solution to the Halting Problem does not exist. The term “Turing degree” is named after him, which refers to a measure of the noncomputability in any solution.

7 Conclusion

The Sum of Cubes question took nearly 70 years to be solved. Interestingly, sometimes making the computer more powerful doesn’t provide the answer. There are still many discoveries to be made when k is increased and in other types of Diophantine equations. Diophantine equations have a long history and remain important in modern applications in astronomy and computer science.

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Numerical Blockade

Aditya Pahuja

This puzzle is a twist on the traditional KenKen puzzle. In a standard KenKen puzzle, the goal is to fill each row and column with the numbers from 1 to n such that the numbers in each region conform with the arithmetic rules of the region (e.g., the numbers in a region labelled “ $5\times$ ” must multiply to 5), where n is the number of cells in each row and column.

In this puzzle, there are four blackened squares called *barriers*. These squares, as their names suggest, serve as barriers, cutting off the rows and columns that they intercept and creating new rows and columns, each of which must be filled with the integers from 1 to n . For example, in the image below, each of the regions outlined in red is a separate row. The row on the left must therefore be filled with the numbers from 1 to 4 and the row on the right must be filled with the numbers from 1 to 7.

2	3	4	5	6	7	8	9	10	11	12	1	
1	2	3	4		1	2	3	4	5	6	7	
3	4	5	6	7	8	9	10	11	12	1	2	

The Puzzle:

19+		27+				480×	60×		2÷	20+	3÷
3-		26+	2904×								
			18+				6×		4-	40×	
3-					180×						11÷
		7+		21+		16+	13+	24×		1200×	
23+	8640×	11									33+
			3		60×		8				
	120×		13+				7×			162×	6
				8+	28×	21+	90×				
1056×										31+	
	4	3÷					120×				
18+			60×			528×				1-	

Solution to Numerical Blockade

Aditya Pahuja

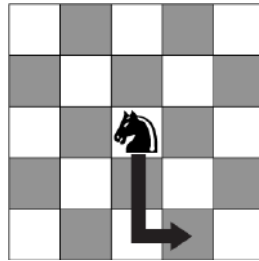
4	5	7	1	10	9	8	2	3	6	11	12
7	1	6	3	11	8	10	5	2	12	9	4
10	9	8	2	12	11	3	6	1	7	4	5
2	3	1		4	3	2	1		3	2	1
5	8	4	2	6	12	7	9	3	1	10	11
3	6	11	1	8	5	9	4	2	10	12	7
9	12	10	3	7	6	1	8	4	11	5	2
11	10	12	4	9	2	5	7	1	8	3	6
1	2	3		2	1	4	3		2	1	3
12	11	2	1	5	4	6	10	3	9	7	8
8	4	9	3	1	7	11	12	2	5	6	10
6	7	5	2	3	10	12	11	1	4	8	9

The Penguin's Tour

Aditya Pahuja

1 Introduction

The *knight's tour* is an ancient puzzle involving a rectangular grid and a knight. A knight is a chess piece that moves two squares horizontally or vertically on the grid and one square in the perpendicular direction, like this:



The goal of the knight's tour is to find a path that the knight can take such that it lands on every single square on the board exactly once. A *closed knight's tour* is a knight's tour that ends with the knight moving to the square that it started on, while an *open knight's tour* is a tour in which the knight does not end on its starting square.

2 The Puzzle

Let a *penguin* be an object that, analogous to a knight, can move three squares horizontally or vertically and two squares in the perpendicular direction. A *closed penguin's tour* would then be a path that a penguin can take such that it lands on every square of the grid exactly once and lands on its original square. Prove that a “closed penguin's tour” is impossible on A) a 7×7 grid and B) a 6×6 grid.



Solution to the Penguin's Tour

Aditya Pahuja

7×7 Grid

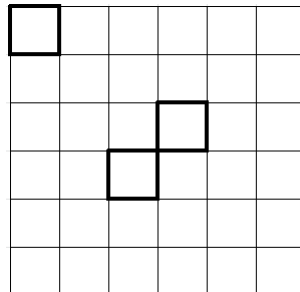
If we color the grid like a chessboard, such that dark squares are only adjacent to light squares and vice versa, we can see that every time a penguin moves, it goes from a square of one color to a square of the other color. This is because the penguin first moves 2 squares in one direction, which lands the penguin on a square of the same color, and then moves 3 squares in the other direction, which lands the penguin on a differently-colored square.

Since the tour alternates between light and dark squares, we can think of a penguin's tour as a polygon where each vertex is a square of the grid and each edge represents a penguin move. We can color each vertex of this polygon to match the color of its square. Since the penguin moves from a light square to a dark square and vice versa, each vertex must be the opposite color from its adjacent vertex. This means that there needs to be an equal number of light and dark squares.

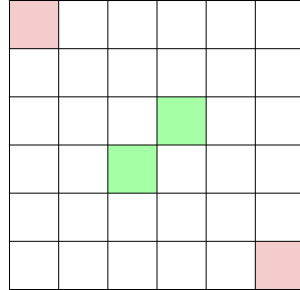
However, in a 7×7 grid, there are 49 squares. This means that it is mathematically impossible for there to be an equal number of light and dark squares, which means that a closed penguin's tour is impossible on a 7×7 grid.

6×6 Grid

On a 6×6 grid, the penguin runs into a problem along its tour: it has no way to access all the corners of the grid! We can see that each corner corresponds to two squares from which the penguin can access that corner:



However, if we look at a pair of diagonally opposing corners, we notice that the access squares for both corners are the same:



Since this coloring of the grid is symmetric, we can say that the other two corners in the diagram are red, and the other two central squares are therefore also green. Using this method of sectioning off the board, we can create three separate cases for the penguin's initial square:

Case 1: Initial square is a corner (red square)

If the penguin starts on a red square, it has to exit onto one of two available green squares. Since the penguin is now on top of a green square, it must go to the only available red square, or else it won't be able to exit that corner later without landing on the same green square twice.

However, once the penguin enters the corner, it must exit onto the only available green square. Since the penguin then has to go to a white square in order to land on all the other squares, it has to leave the access square, which means that it has no avenue back to its initial square, which violates the conditions of a closed penguin's tour and therefore makes the tour impossible.

Case 2: Initial square is an access square (green square)

If the penguin instead starts on a green square, it immediately needs to go to a corner square, since going to a non-corner square will prevent the penguin from reaching both corners because the penguin is unable to re-enter its initial square except to close the tour, therefore preventing it from reaching both corners.

Since the penguin needs to go to a corner, it then has to exit through the other access square. In order to reach the diagonally-opposite corner, it has to go to the corner immediately after entering that access square, since not doing so would lock off that corner. However, upon entering the corner, both access squares have already been used, resulting in the penguin being unable to exit, which means that regardless of what happens, the penguin is unable to reach every square on the board if it starts on an access square.

Case 3: Initial square is neither an access square nor a corner (white square)

If the penguin starts on a white square, then the only way for it to get to a corner square is to get to an access square first. Going to an access square without reaching any corners first has already been proven in Case 2 to not work, which means that starting on a white square is futile.

“Ketch up” on the KenKen

Edward Wu

Another day, another KenKen. You print out a copy and settle down with your hamburger (or vegan burger) with ketchup and whatever other toppings you fancy. As you take your first bite, *SPLAT!* You freeze, worried about what you will see when your eyes open, and to your horror, ketchup covers some of the clues of your KenKen, seeping through the paper so the clues are illegible from the other side as well. You don’t want to print another copy because that’s a waste of paper and ink, and you are still a dedicated environmentalist. See if you can finish it! (P.S. You can.)

48 ×			27 +		14 +			
3 ÷				11 +		18 ×		17 +
12 ×	8 +	840 ×				18 +		
			30 +					
17 +				?		25 +		
70 ×	??		54 ×		432 ×			8 +
				4 −			??	
6 −	2 −	??		192 ×	??			??

Solution to “Ketch up” on the KenKen

Edward Wu

NOTE: For this solution, we call the columns A through I from left to right and the rows 1 through 9 from top to bottom. Then the squares $B3$ and $B4$ make up the $8+$ clue in the second column. A cell is defined to be the intersection of a row and column.

The solution has been broken into sizable parts. Read at your own pace.

When asked to “recall,” the author has supplemented subscripts indicating from which previous part this detail has been proved. This is a subscript. Superscripts denote footnotes.

Images have been provided at certain points, for the reader’s convenience.

Part I

KenKens are an interesting type of puzzle because you need somewhere to start. This probably is $\frac{1}{4}$ of the puzzle, because you have no clues when you begin, and you have to search for where to start. It’s easy to continue working around the grid once you have your first number. Although the six ? clues give fewer clues for the entire puzzle, sometimes not all the clues are necessary, which is why the puzzle is still completable despite the absence of six clues. In fact, this will help determine where to start, as there are now six less clues to consider.

Not sure if there is a second place to begin, but the place where I began is in Row 2. Notice in Row 2 there are a $3\div$ and a $18\times$ clue. Both of them have a limited number of possibilities. The $3\div$ clue can be $(1, 3)$, $(2, 6)$, or $(3, 9)$. The $18\times$ clue can be $(2, 9)$ or $(3, 6)$. Notice the contradiction?

If the $18\times$ clue is $(3, 6)$, then the $3\div$ clue has no possibilities because neither 3 nor 6 can be in the $3\div$ clue. Then the $18\times$ clue must be $(2, 9)$ in some order. Since the 2 and 9 are both used, the $3\div$ must be $(1, 3)$ in some order.

This is the first major realization of the puzzle.

Part II

The next point of attention is the $48\times$ clue in Row 1. Because the entire clue is in Row 1, there cannot be any repeats. The only possibilities are $(1, 6, 8)$, $(2, 3, 8)$, $(2, 4, 6)$. You can prove these are all the possibilities because there must be a 6 or an 8. The next biggest factor of 48 is 4, with the largest product you can make with 4 being the largest of three factors is $2 \times 3 \times 4 = 24$.

This may not seem very important, but the $14+$ clue that spans four cells is. The average of these four distinct digits is 3.5, which is pretty concerning since the average is expected to be 5. How many ways can you make 14? $(1, 2, 3, 8)$, $(1, 2, 4, 7)$, $(1, 2, 5, 6)$, $(1, 3, 4, 6)$, $(2, 3, 4, 5)$ are all the possibilities. Notice that only one of these combinations does not have a 1, and only one does not have a 2. Use your skills from Part I to determine which is the only possible outcome from this row.

Did you get it? The single possibility for $14+$ without a 2 is $(1, 3, 4, 6)$. If the $48\times$ clue is $(2, 3, 8)$ or $(2, 4, 6)$ then the 2 will not be in the $14+$. Then the $14+$ is $(1, 3, 4, 6)$. However, no matter which of the two triplets for $48\times$ we use, the 3 or 4 will be used. This is a contradiction. Thus, the $48\times$ is $(1, 6, 8)$ and the $14+$ is $(2, 3, 4, 5)$.

This is the second major realization of the puzzle.

Part III

Now consider Rows 1 and 2 together. So far, we have deduced that $48\times$ is $(1, 6, 8)$, the $14+$ is $(2, 3, 4, 5)$, the $3\div$ is $(1, 3)$, and the $18\times$ is $(2, 9)$. Because each digit from 1 to 9 appears once in each row, the sum of the numbers in each row is 45. Two rows add to 90. Thus, in the first two rows, the digits in $48\times$ add to 15. The digits in $3\div$ add to 4. The digits in $18\times$ add to 11. Thus, cell $I2$ must be $90 - 15 - 27 - 14 - 4 - 11 - 11 = 8$.

This is the third major realization of the puzzle.

Part IV

You probably already noticed the $70\times$ that spans Rows 6 and 7 must be made up by $(1, 2, 5, 7)$ because 10 is not an option as the grid is 9×9 .

Confined in Columns A through C , there are 3 clues that all use 1. Since each digit appears once in each column, each digit must appear exactly three times in any three columns. Thus, the 1 digit appears three times in Columns A through C , and we have already identified in which three clues they will be used. This means there cannot be any other 1s in Columns A through C .

	A	B	C	D	E	F	G	H	I
1				27+					
	1 6 8	1 6 8	1 6 8			2 3 4 5	2 3 4 5	2 3 4 5	2 3 4 5
2					11+				8
	1 3	1 3					2 9	2 9	
3	12x	8+	840x				18+		
4				30+					
5	17+				?		25+		
6		??		54x		432x			8+
	1 2 5 7								
7					4-			??	
	1 2 5 7	1 2 5 7	1 2 5 7						
8	6-	2-	??		192x	??			??
9									

This next part is hard to see. Consider the 6− clue in Column A. There are 3 options for a 6−: (1, 7), (2, 8), (3, 9). Because we know there cannot be a 1, the (1, 7) case is gone. Consider the (3, 9) case. This means that the 12× in Column A is (2, 6) because the 3 for (3, 4) is taken. A2 is a 1 because the 3 in Column A is used in 6−. Recall that 48× in Row 1 is (1, 6, 8) 2. In Column A, cell A2 is the 1 and the 6 is used in 12×. Then cell A1 is 8 and cells A2 and A3 are some permutation of (1, 6).

Because cell A2 is 1, cell B2 is 3. Now consider the 8+ in Column B. There are also three choices for this: (1, 7), (2, 6), (3, 5). Because we have used up all the 1s, (1, 7) is eliminated. We also used the 3 in the 3÷ so the 8+ must be (2, 6). Then cell A2 is 1 because 6 is used in the 8+.

This is the last part of this hypothetical. Recall that 70× is (1, 2, 5, 7)₄. Looking back at Column A, the 1 is in cell A2 and the 2 is in the 12× so cells A6 and A7 are 5 and 7 in some order. Thus 1 and 2 are in cells B7 and C7. However, in Column B, the 1 is in cell B1 and the 2 is used in 8+. Thus, there is no possible digit for cell B7, and this entire hypothetical is impossible.

Thus, the 6− that started all this in Column A cannot be (3, 9) so it must be (2, 8).

This is the fourth major realization of the puzzle.

Part V

We will now use what we just learned. Because the 6− is (2, 8), the 12× cannot have a 2, so it must be (3, 4). This means the 3 in Column A is used in 12×. Thus, cell A2 is a 1, and cell B2 is a 3.

Inspect Column A. We have used the 1 in cell A2, the 2 and 8 in cells A8 and A9, and the 3 and 4 in cells A3 and A4. The remaining cells (A1, A5, A6, A7) are filled with the four remaining digits (5, 6, 7, 9). Notice that 9 cannot be in 12× or 70×. Thus, 9 must be in the 17+, so cell A5 is a 9. Now that cell A5 is

taken, cell $A1$ must be the 6.

Recall the $8+$ in Column B_4 . We know there is no 1, and the 3 is used in cell $B2$ so $8+$ must be $(2, 6)$ because $(1, 7)$ and $(3, 5)$ are impossible. Thus, cell $B7$ is a 1, and cell $C7$ is a 2. Then in $48\times$ in Row 1, cell $B1$ is a 8, and cell $C1$ is a 1.

Notice that in Column B we have found the precise locations of the 1, 3, 8. We also know that the $8+$ is made up by $(2, 6)$. The remaining four cells $(B5, B6, B8, B9)$ are filled by the remaining four digits $(4, 5, 7, 9)$.

Consider the $17+$ in Row 5. Because cell $A5$ is a 9, the remaining three cells $(B5, C5, D5)$ add up to 8. Simply considering the remaining possibilities for Column B restricts cell $B5$ to be 4, 5, 7, or 9. It of course cannot be 9. It also cannot be 7 because the two remaining cells $C5$ and $D5$ would have a sum of 1. Furthermore, it cannot be 5 because then cells $C5$ and $D5$ have a sum of 3. Although it is possible to achieve a sum of 3 in two cells, it would have to be made up of 1 and 2, both of which are used in Column C already. Thus, cell $B5$ is a 4.

Then $C5 + D5 = 4$, which has one solution, $(1, 3)$. The 1 in Column C is in $C1$, so cell $C5$ is a 3, and cell $D5$ is the 1.

	A	B	C	D	E	F	G	H	I
1	6	8	1	27+		2 3 4 5	2 3 4 5	2 3 4 5	2 3 4 5
2	1	3			11+		2 9	2 9	8
3			840x				18+		
4				30+					
5	9	4	3	1	?		25+		
6		??		54x		432x			8+
7			1	2	4-			??	
8		2-	??		192x	??			??
9									

This is the fifth major realization of the puzzle.

Part VI

Consider the $54\times$ clue that spans Columns D and E . Notice that because of its L shape, it allows you to repeat a digit twice in cell $E6$. What are the possibilities for a $54\times$ in four cells in this shape?: $(1, 1, 6, 9)$, $(1, 2, 3, 9)$, and $(1, 3, 3, 6)$. Notice all of these options force a 1. Then there must be a 1 in the $54\times$. There is already a 1 in cell $D5$, so the 1 must go in cell $E6$.

Out of the three quadruples, only one can exist. $(1, 1, 6, 9)$ is eliminated because there cannot be a second

1 in Column D . $(1, 3, 3, 6)$ is also eliminated since 1 occupies cell $E6$, forcing both 3s into Column D . Thus, $54\times = (1, 2, 3, 9)$.

This is the sixth major realization of the puzzle.

Part VII

Return to Row 1. The digits in $48\times$ are 1,6,8. The digits in $14+$ are 2,3,4,5. Thus, cells $D1$ and $E1$ are 7 and 9. We know there is a 9 in Column D in the $54\times$ clue. Thus, cell $D1$ is 7 and cell $E1$ is 9.

Consider Row 3. Where can the 9 go? Cell $A3$ is a 3 or 4; cell $B3$ is 2 or 6; 9 is not a factor of 840 so it cannot be cells $C3, D3, E3, F3$; with 8 in cell $I2$, $I3 + I4$ add to 9, so cell $I3$ cannot be 9. Therefore, 9 is in cell $G3$ or cell $H3$.

We have deduced the $18\times$ in Row 2 is $(2, 9)_1$. Thus, there are already two 9s in Columns G and H , so there cannot be any other 9s in these two columns.

Consider Row 4. Where can the 9 go? We just deduced that 9 cannot be in Columns G and H . Using similar logic as for Row 3, the 9 cannot be in cells $A4, B4, C4$ or $I4$. Then the only choices left for the 9 in Row 4 are in cells $D4, E4$, and $F4$.

Columns D and E have 9s already. Remember that the position of the 9 in column D is not precisely known, but we know it is in the $54\times$ clue 6. Since these two columns have 9s, the 9 in row 4 must be in cell $F4$.

You can use this same logic to find a 9 in one of the remaining columns where the position of the 9 is not precisely known (B, C, D, G, H, I). Try to find another 9. The answer is at the end of the article¹.

This is the seventh major realization of the puzzle.

Part VIII

Consider the $840\times$ that spans Rows 3 and 4. The prime factorization of 840 is $840 = 2^3 \cdot 3 \cdot 5 \cdot 7$. Notice that two of these five cells must be occupied by 5 and 7. The remaining three cells are occupied by digits that multiply to 24. Notice that because the $840\times$ spans two rows, there can be two of a number. However, this number cannot be 1, 2, or 3 because these three digits already occupy positions in Column C , where the repeated digit would appear.

A $24\times$ spans three cells and allows one repeat which is greater than 3, so it can be filled with $(1, 3, 8)$, $(1, 4, 6)$, or $(2, 3, 4)$. Call the sum of the digits of whichever triplet is the correct triplet X . Let Y be the value of cell $F5$.

In Rows 3 and 4, the sum of these digits is $2 \cdot 45 = 90$. In Column A we have a sum of 7 across Rows 3 and 4. In Column B we have a sum of 8. There is guaranteed to be a 5 and a 7 in the $840\times$. Thus, we can write $7 + 8 + (12 + X) + (30 - Y) + 18 + (17 - 8) = 90$ which simplifies to $X - Y = 6$.

Consider the possible values for X . If the triplet is (1,3,8), $X = 12$. If it is (1,4,6), $X = 11$. If it is (2,3,4), $X = 9$. The respective values for Y that solve $X - Y = 6$ are $Y = 6$, $Y = 5$, and $Y = 3$.

Y cannot be 3 because 3 is already in Row 5 in cell $C5$. The other two possibilities are still valid, however. This means that the corresponding triple (2,3,4) cannot be the one for $840\times$. Thus, the $840\times$ must be (1,3,5,7,8) or (1,4,5,6,7). Either way, there is a 1. Notice Columns C, D, E all have 1s placed. Thus, the 1 is in cell $F3$.

Now consider the entire Column I . Where can the 1 go? The remaining three rows without 1s are Rows 4, 8, and 9. However, the 1 cannot be in Row 4 because that would force 8 into cell $I3$ to complete $17+$, but cell $I2$ is already an 8. Therefore, cell $I8$ or cell $I9$ is 1. The 9 in Column I cannot be in cell $I1$ because there is a 9 in cell $E1$, cell $I2$ is taken up by an 8, cells $I3$ and $I4$ add to 9 so neither can be the 9, cell $I5$ cannot be a 9 because 9 is in cell $A5$, 9 cannot be in cell $I6$ or cell $I7$ because the two cells add to 8. Thus, 9 is also in cell $I8$ or cell $I9$.

Because the 1 and 9 take up $I8$ and $I9$ in column I , the remaining 7 digits take up the other 7 spots. The digits in Column I add to 45. $I2 + I3 + I4 = 17$, $I6 + I7 = 8$, $I8$ and $I9$ are 1 and 9 in some order. Thus, $I1 + I5 = 45 - 17 - 8 - 1 - 9 = 10$.

Consider the remaining free cells in Row 5. We have narrowed down the possibilities for cell $F5$ to either 5 or 6. Notice that the $25+$ cannot have any 9s because cell $A5$ is 9. Furthermore, we have deduced that columns G and H cannot have any more 9s. Thus, the max digit for any cell in the $25+$ is 8. If the 2 in Row 5 is placed in the $25+$, then the three remaining cells add to 23.

Because there are no 9s, the only way to add to 23 is (7,8,8). Because 8 is in $I2$, the 7 goes in cell $H5$, the 8s in cells $G5$ and $H6$, and the 2 in cell $I5$. Recall that $I1 + I5 = 10_8$. This makes cell $I1 = 8$, but 8 is in cell $I2$, so this is a contradiction. Thus, the 2 is not in the $25+$, so it is in cell $E5$.

	A	B	C	D	E	F	G	H	I
1	6	8	1	7	9	2 3 4 5	2 3 4 5	2 3 4 5	2 3 4 5
2	1	3			11+		2 9	2 9	8
3			840x			1	18+		
4	3 4	2 6		30+		9			
5	9	4	3	1	2	5 6	25+		
6		??			1	432x			8+
7		5 7	1	2	9	4-		??	
8		2-	??		192x	??			1 9
9		2 8							1 9

This is the eighth major realization of the puzzle.

Part IX

At this point, there are many cells that are down to two choices. We can pick one of these and see which of the choices will lead to a contradiction.

For simplicity's sake, the cell in question is cell $H6$.

We have already proved that cell $F5$ is a 5 or a 6. Then the remaining cells in Row 5 are (6,7,8) or (5,7,8), which will have sums of 21 and 20, respectively. This would make cell $H6$ either a 4 or a 5.

Which would be more useful to test first? In Column A , we also have a 5 that could potentially be in Row 6. Thus, we should let cell $H6$ be 5.

Immediately, this makes cell $A6$ be 7 and cell $A7$ be 5. Notice in Column B that the 5,7,9 go in cells $B6$, $B8$, and $B9$. Since cell $A6$ is 7 and we assumed cell $H6$ is 5, cell $B6$ must be 9. Furthermore, in Column I , the three choices for an 8+ clue that spans two cells (1,7),(2,6),(3,5) has been reduced to one. (1,7) and (3,5) are both impossible because 1s and 5s already occur in both Rows 6 and 7. Because cell $C7$ is a 2, cell $I6$ must be the 2, and cell $I7$ must be the 6. Consequently, cell $D6$ is a 3, and cell $D8$ is a 2.

Let's consider the 432x clue with $432 = 2^4 \cdot 3^3$. We have already proved that there are no more 9s in Columns G and H . There is also a 9 in Column F in cell $F4$. Then there must be no 9s in the 432x. Then we need two 3s and a 6 or two 6s and a 3. Since cell $D6$ is a 3, cell $F6$ must be a 6 because the number in this cell is the only digit in this clue that can repeat.

From our original assumption that cell $H6$ is a 5, we know the sum of the digits in cells $G5$, $H5$, and $I5$ is 20. Then out of (5,6,7,8) that are still remaining in Row 5, the 6 is not used. Then the 6 must be in cell $F5$. However, we just found that cell $F6$ is a 6. There cannot be two 6s in the same column, so this is a contradiction, and our original assumption was incorrect.

Thus cell $H6$ must be a 4 and cell $F5$ must be a 5.

This is the ninth major realization of the puzzle.²

Part X

One feature that I love about KenKens is that although you may be concentrating on one clue or one part of the board, it will have influence over cells somewhere else.³

Where was cell $H6$ useful? It helped find cell $F5$. Where was cell $F5$ useful? If you recall, cell $F5$ was our value of Y_8 . Now we know that X is 11, so the digits in the 840x clue are (1,4,5,6,7).

Since the 840x clue is (1,4,5,6,7), then there is no 8. Notice in Row 3 that we have not found a location for the 8. We have just knocked out cells $C3$, $D3$, and $E3$ as possible habitats for the 8. The 8 also definitely cannot be in cells $A3$ or $B3$, and there is an 8 in cell $I2$. Then the 8 must be in cell $G3$ or cell $H3$, both of which are in a 18+ clue.

Recall that the 9 in Row 3 is also in cell $G3$ or cell $H3$ ₇. Then the sum of these two cells must be $8+9=17$. Then we know for sure that cell $H4$ is a 1.

Notice that in Rows 3 and 4, because the $840\times$ clue is $(1,4,5,6,7)$, we are using both 4s and both 6s in these two rows. Consider cell $I3$. Recall that the digits in cells $I3$ and $I4$ add to 9 ₈. Since we used both 4s and 6s in these two rows, neither digit can appear in this clue, so the $17+$ must be $(2,7,8)$.

Finishing up Column I , we still have digits 3,4,5, and 6. Only 6 can go in cell $I5$ so we can fill that in. Also notice that the only way to satisfy $8+$ is $(3,5)$. Then cell $I1$ must be the 4.

	A	B	C	D	E	F	G	H	I	
1	6	8	1	7	9		2 3	2 3 5	2 3 5	4
2	1	3			11+			2 9	2 9	8
3			840x			1		8 9	8 9	2 7
4				30+		9		1		2 7
5	9	4	3	1	2	5		7 8	7 8	6
6		??			2 3	1	432x		4	3 5
7		5 7	1	2	9	4-			??	3 5
8		2-	??		2 3	192x	??			1 9
9		2 8								1 9

Part XI

Return to Row 3. Consider the digit 3. The only possible spot for the 3 is in cell $A3$ because $840\times$ is taken up by $(1,4,5,6,7)$ ₁₀. Cell $A4$ must be a 4.

Recall the principle that in two columns, each digit appears exactly twice and no more than *twice* _{4,7}. Then in Columns G and H , the digit 8 appears no more than twice. We proved there must be an 8 in cell $G3$ or cell $H3$ ₁₀. We also know that cell $G5$ or cell $H5$ is an 8 because these two are the only two empty spots left in the row. Then there cannot be any more 8s. There also cannot be any more 9s.

Consider the $432\times$ clue where $432 = 2^4 \cdot 3^3$. There is a 9 in cell $F4$, and we just proved there cannot be any more 9s in Column G . So we need to achieve 3^3 without 9s. Then there must be either two 3s and a 6 or two 6s and a 3. Either way, whichever digit is repeated must appear in cell $F6$. What are the possibilities? If the repeat is a 3, we have $(3,3,6,1,8)$ and $(3,3,6,2,4)$, but we can eliminate the first because we just proved there cannot be an 8. If the repeat is a 6, we have $(3,6,6,1,4)$. Therefore, there is confirmed to be a 3,4, and 6 in Column G within the bounds of the $432\times$ clue.

Now consider Row 4. We have already determined the positions of the 1,4,9. We also know that cell $F5$ is a 5. What digits are left for this row?: 2,3,5,6,7, and 8. Notice that within Rows 3 and 4, there will be a 2 in Column B and there will be a 2 in Column I . Thus, there cannot be a 2 in the 30+ clue. Similarly, we can eliminate the 6 and 7 because one is guaranteed in Columns B and I , respectively, and the other is guaranteed in the $840\times$ clue. Thus, the remaining numbers for the 30+ clue are 3,5,8.

There are already two 8s in Columns G and H so cell $G4$ cannot be an 8. Furthermore, we just proved there must be a 3 in the $432\times$ clue in Column G . Therefore, cell $G4$ is a 5. Similarly, because there is confirmed to be a 3 in the $54\times$ clue in Column D , cell $D4$ is the 8, and cell $E4$ is the 3. We are almost done!

Now that cell $G4$ is confirmed to be 5, let's inspect where the 7 in Column G can go. It cannot be in cell $G1$ because $14+$ is (2,3,4,5). The 7 cannot be in cell $G2$ because $18\times$ is (2,9). It cannot be in cell $G3$ because $18+$ is (1,8,9). Cell $G4$ is taken. The 7 does not divide 432, so it cannot be in cells $G6, G7, G8$, or $G9$. Therefore, cell $G5$ is a 7.

Consequently, cell $H5$ is a 8, cell $H3$ is a 9, cell $H2$ is a 2, cell $G2$ is a 9, and cell $G3$ is a 8.

Furthermore, because 5s appear in Columns F and G , cell $H1$ must be the 5 in the $14+$ clue.

We also proved that within cells $G6, G7, G8$, and $G9$, there must be a 3,4, and 6. Then cell $G1$ cannot be 3, so it must be 2. Then cell $F1$ is a 3. But then cell $F6$ cannot be a 3, so it must be a 6. Aren't KenKens nice? Clues loop back to solve each other!

The next list can be verified easily (with the clues you've obtained from reading 7 pages of solution up to this point!)

Cell $G6$ is a 3, cell $I6$ is 5, cell $I7$ is 3, cell $A6$ is 7, cell $A7$ is 5, cell $D6$ is 2, cell $D8$ is 3, cell $B6$ is 9, and cell $C6$ is 8.

Since cell $G6$ is a 3, cell $F6$ cannot be a 3, so cell $F6$ must be a 6.

In Row 7, the 4- clue must be (4,8) because (1,2,3,5) are all taken in the row. Then in Row 7, the 7 must be in cell $H7$ because 7 does not divide 432. This means the 6 is in cell $G7$.

We will return where we started: Row 2. There, the $11+$ clue is left to (4,7) or (5,6). Notice that in column F we have already used both 5 and 6 in cells $F5$ and $F6$ (coincidence?) respectively. This means that the $11+$ clue must be (4,7).

Our favorite conclusion: in two columns, each digit appears exactly twice, so 4,7,11, will be reused in Columns E and F . We know there is a 4 in Row 2 and a 4 in Row 7. Then there is no 4 in Rows 8 or 9 in Columns E and F . How can we satisfy the $192\times$ clue? $192 = 2^6 \cdot 3$. Because we already used both 3s in Columns D and E , there must be a 6. Then the other two numbers multiply to $25=32$, so they must be 4 and 8. Then there must be a 4 in the $192\times$ clue.

We just explained that there cannot be any more 4s in Columns E and F . Then cells $E8$ and $E9$ cannot be the 4. The 4 must be in cell $D9$. Because the 4 is in cell $D9$, cells $E8$ and $E9$ must be 6 and 8. This

means that cell $E7$ cannot be 8, so cell $E7$ is 4.

More simple solving because you are a KenKen pro!!

Cell $F7$ is 8, cell $G8$ is 4, cell $G9$ is 1, cell $I8$ is 1, cell $I9$ is 9, cell $E2$ is 7, cell $F2$ is 4, cell $E3$ is 5, and cell $C3$ is 4.

	A	B	C	D	E	F	G	H	I
1	6	8	1	7	9	3	2	5	4
2	1	3			7	4	9	2	8
3	3	_{2 6}	4		5	1	8	9	_{2 7}
4	4	_{2 6}	_{6 7}	8	3	9	5	1	_{2 7}
5	9	4	3	1	2	5	7	8	6
6	7	??		2	1	6	3	4	5
7	5	1	2	9	4	8	6	7	3
8	_{2 8}	₂₋	??	3	_{6 8}	??	4		1
9	_{2 8}			4	_{6 8}		1		9

Part XII

From here, we have used all known clues. Now is the time to bust out our pro Sudoku skills!⁴

Column H is still missing a 3, but it can only go in cell $H8$ or cell $H9$. In Row 8, there is a 3 occupying cell $D8$. Therefore the 3 in Column H must go in cell $H9$. The last digit missing in Column H , a 6, goes in cell $H8$.

Likewise, cell $E9$ is a 6, and cell $E8$ is a 8. Cell $A9$ is a 9, and cell $A8$ is a 2. Cell $F9$ is a 2, and cell $F8$ is a 7. Cell $B9$ is a 7, and cell $B8$ is a 5. Just take a moment to admire this chain of events.

Row 9 is missing a 5 so it must go in the last empty cell, cell $C9$. Similarly, cell $C8$ is a 9.

From here, I must depart. As my last words to you, I know you are destined to go on and complete more great puzzles. Remember your training! Go you!

Part XIII

The remainder of the puzzle is left as an exercise to the reader.

	A	B	C	D	E	F	G	H	I
1	6	8	1	7	9	3	2	5	4
2	1	3			7	4	9	2	8
3	3	_{2 6}	4		5	1	8	9	_{2 7}
4	4	_{2 6}	_{6 7}	8	3	9	5	1	_{2 7}
5	9	4	3	1	2	5	7	8	6
6	7	??		2	1	6	3	4	5
7	5	1	2	9	4	8	6	7	3
8	2	5	9	3	8	7	4	6	1
9	8	7	5	4	6	2	1	3	9

Notes

1) Answer: The 9 in Column *D* is in cell *D7*. In Row 7, there cannot be a 9 in cells *A7* or *I7* because they make the clue impossible to achieve. The digits 1 and 2 occupy cells *B7* and *C7*. There are 9s in Columns *E, F, G, H* that are confirmed to not be in Row 7. Thus the 9 is in cell *D7*.

2) Although it may seem tedious and stumbled upon by chance, sometimes you will be forced to bash one option until you reach a contradiction. What's important is that we have the mindset that is willing to bash. Bashing is not a terrible technique. Furthermore, we have already limited the options of the cell from 9 down to 2, which is already tremendous work.

3) If you do Kakuro, you may notice that if two clues do not intersect, there is basically no relationship between the two. This is not the case with KenKens.

4) Sudoku is a game similar to KenKen. Usually there is a 9×9 grid separated into nine 3×3 grids. In each row, column, and 3×3 grid, each digit from 1 to 9 appears once. There are no arithmetic clues.

Tangrams

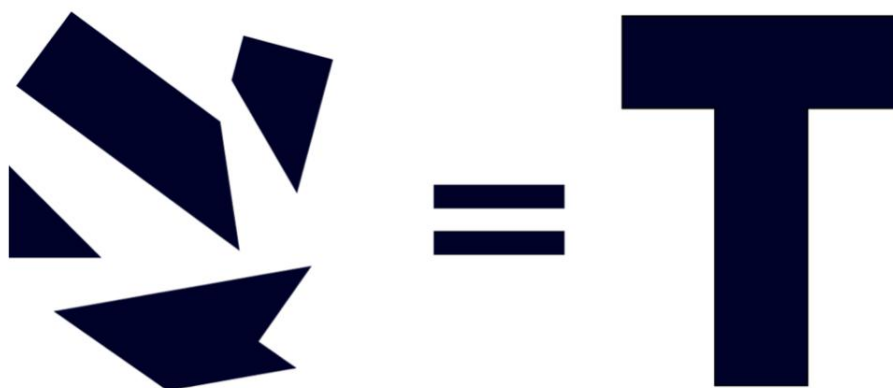
Nathan Kuo

1 Introduction

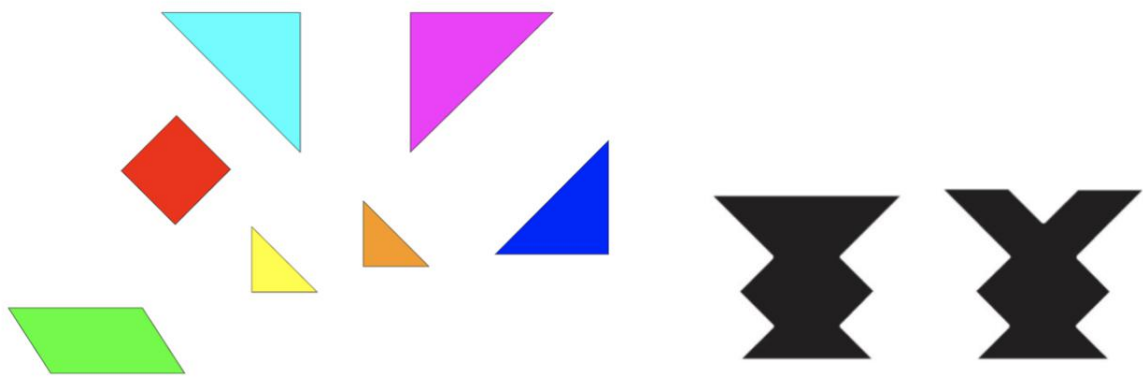
Tangrams are a classic puzzle in which seven flat polygons come together to form unique shapes. The seven polygon limit will not be followed as that limits some possibilities of the shapes. A tangram paradox is a dissection fallacy: two figures which are made with the same set of pieces, one of which appears to be a subset of the other. Here, a 10-sided polygon can be made out of seven pieces, but the second, even though it looks as if a triangle has been taken out, still uses the same pieces. In addition, the classic T tangram is here.

2 The Puzzles

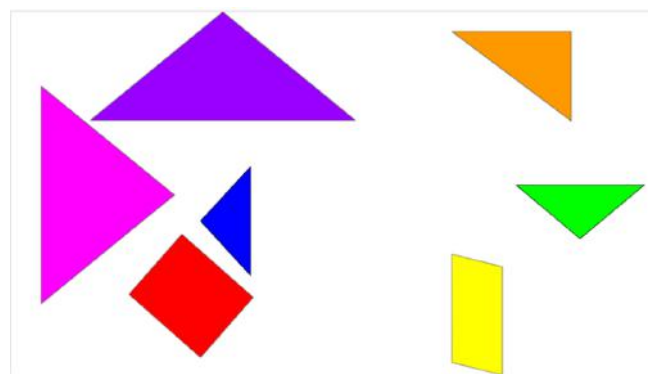
Using these four pieces, make the T:



Using the seven colored pieces, make both black shapes pictured below:



Using the seven colored pieces, make both figures pictured below:



The Tower That Isn't a Hanoi Tower

Nathan Kuo

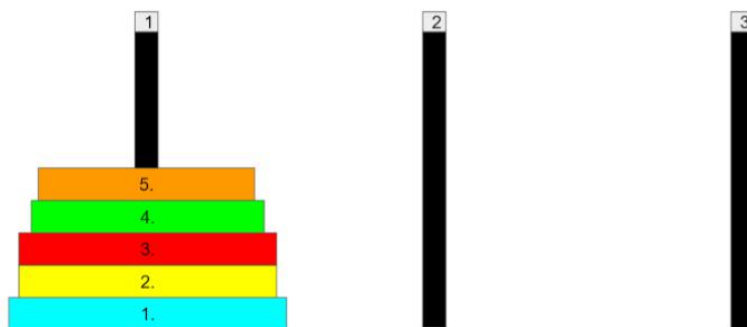
This tower is not a Hanoi Tower. Trust me.

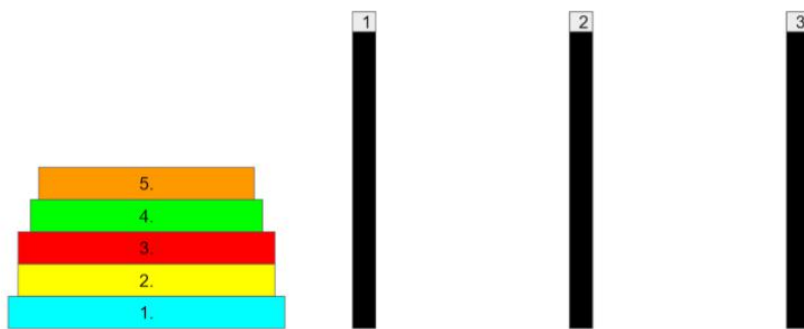
This puzzle, based off of a Hanoi Tower, consists of three poles and a tower of disks of different sizes that are initially stacked to form a cone. The objective of this puzzle is to move the entire stack to another rod while following these guidelines:

- Only one disk can be moved at a time.
- Each move consists of taking the upper disc from one of the stacks and placing it on top of another disc or an empty rod.
- The number of disks on pole 3 cannot exceed 3 until the final construction of the tower begins.

Two disks are the same size to allow for more variability, or completely mess up what you were planning to do.

Puzzles

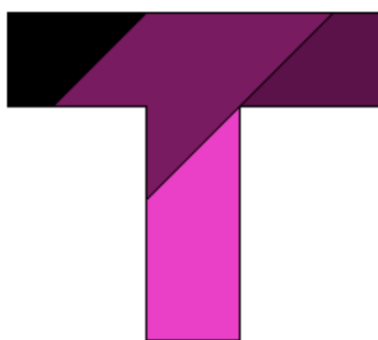


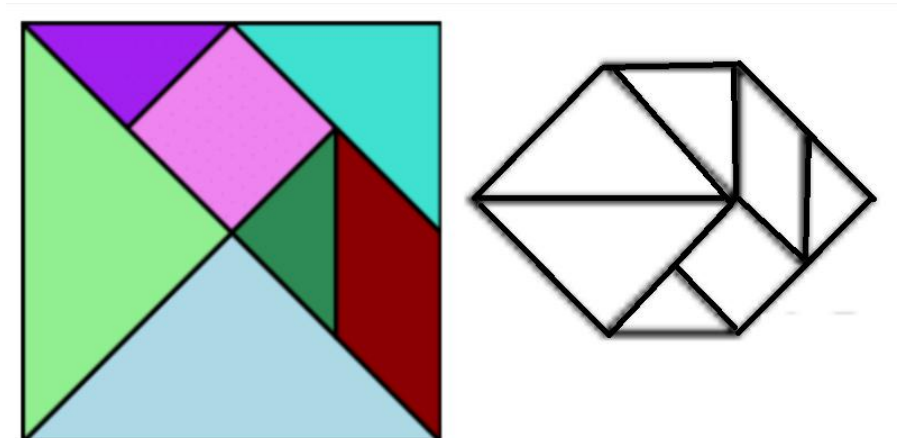


Solutions to Tangram and “Hanoi Tower” Puzzles

Nathan Kuo

Tangram Solutions





“Hanoi Tower” Solutions

1. Move disk 5 to pole 3.
2. Move disk 4 to pole 2.
3. Move disk 5 to pole 2 on top of disk 4.
4. Move disk 3 to pole 3.
5. Move disk 2 to pole 3 on top of disk 3.
6. Move disk 5 to pole 3 on top of disk 2.
7. Move disk 4 to pole 1 on top of disk 1.
8. Move disk 5 to pole 1 on top of disk 4.
9. Move disk 2 to pole 2.
10. Move disk 3 to pole 2 on top of disk 2.
11. Move disk 5 to pole 3.
12. Move disk 4 to pole 2 on top of disk 3.
13. Move disk 5 to pole 2 on top of disk 4.
14. Move disk 1 to pole 3.
15. Move disk 5 to pole 3 on top of disk 1.
16. Move disk 4 to pole 1.

17. Move disk 5 to pole 1 on top of disk 4.
18. Move disk 3 to pole 3 on top of disk 1.
19. Move disk 2 to pole 3 on top of disk 3.
20. Move disk 5 to pole 2.
21. Move disk 4 to pole 3 on top of disk 2.
22. Move disk 5 to pole 3 on top of disk 4.

The 2021 Puzzle

Andrew Juang

The 2021 Puzzle

Use all of the digits of 2021 (2,0,2,1) to write mathematical expressions that give results for the numbers 1-27.

Rules:

You can use the arithmetic operations (addition, subtraction, multiplication, division), and you can use factorial and exponentiation. You are allowed to concatenate digits (e.g., $41=20+21$), but not allowed to do something like $(2+1)0 = 30$. You may only use each of the digits once. Try to find as many solutions as you can.

Example: To create 1, you can do $2*0*2+1 = 1$.

1 =
2 =
3 =
4 =
5 =
6 =
7 =
8 =
9 =
10 =
11 =
12 =
13 =
14 =
15 =
16 =
17 =
18 =
19 =
20 =
21 =

$$22 =$$

$$23 =$$

$$24 =$$

$$25 =$$

$$26 =$$

$$27 =$$

Some Solutions to The 2021 Puzzle

Andrew Juang

Some possible answers (out of multiple answers):

$$1 = 21 - 20$$

$$2 = 0 * 2 * 1 + 2$$

$$3 = 2 + 1 + 0 * 2$$

$$4 = 2 + 2 + 0 * 1$$

$$5 = 2 + 0 + 2 + 1$$

$$6 = 2 + 0! + 2 + 1$$

$$7 = 2^{(2+1)} - 0!$$

$$8 = 2^{(2+1)} + 0$$

$$9 = (2 + 1)^2 + 0$$

$$10 = (2 + 1)^2 + 0!$$

$$11 = 12 - 2 + 0!$$

$$12 = 12 + 0 * 2$$

$$13 = 12 + 2 - 0!$$

$$14 = 12 + 2 + 0$$

$$15 = 12 + 2 + 0!$$

$$16 = (2 + 1 + 0!)^2$$

$$17 = 20 - 2 - 1$$

$$18 = 21 - 2 - 0!$$

$$19 = 21 - 2 + 0$$

$$20 = 21 - 2 + 0!$$

$$21 = 22 - 1 + 0$$

$$22 = 21 + 2^0$$

$$23 = 21 + 2 + 0$$

$$24 = (2 + 2)! + 0 * 1$$

$$25 = (2 + 2)! + 1 + 0$$

$$26 = (2 + 2)! + 1 + 0!$$

$$27 = (2 + 0!)^{(2+1)}$$

Questions and Answers with Professor Lisa Randall, Class of 1980



Professor Lisa Randall researches theoretical particle physics and cosmology at Harvard University, where she is the Frank B. Baird, Jr. Professor of Science on the physics faculty. She graduated from Stuyvesant High School in 1980, where she contributed to the Math Team and Math Survey. Currently a member of the American Academy of Arts and Sciences, the National Academy of Sciences, and the American Philosophical Society, she is the recipient of many awards and honorary degrees. Professor Randall was included in 2007's list of TIME Magazine's "100 Most Influential People" and was among Esquire magazine's "75 Most Influential People of the 21st Century." Professor Randall's two books, Warped Passages (2005) and Knocking on Heaven's Door (2011) are New York Times bestsellers and are featured on their lists of 100 Notable Books. She is also known for her contributions to the Randall-Sundrum model and for her research in elementary particles and fundamental forces (Harper Collins Publishers).

Professor Randall's answers are taken from an e-mail interview conducted by Josephine Lee and have been edited for clarity.

1) Why do you enjoy physics so much?

I like making connections to apparently disparate phenomena. It is very satisfying. I also like problem solving. And of course the opportunity to think about the world.

2) Who or what inspired your passion for math/science and physics in particular?

I always liked math. Of course my first real introduction to physics was in high school, and Mr. Baumel was a great teacher! Solving physics problems and learning how things operate was also very satisfying.

3) What do you remember about contributing to *Math Survey*? How did it impact you?

I remember the Math Team much more. It impacted me by making me wake up insanely early! (I had an hour and twenty minute commute each way). But with both Math Team and *Math Survey*, I loved the opportunity to be with such incredibly smart students.

4) Were there any specific teachers, people, or memories of Stuyvesant that have had a significant impact on you? Looking back, what would you tell your 16-year-old self?

I guess I already answered that... But mostly I also remembered the celebration of quirkiness. In Fresh Meadows, where I grew up, you were condemned for being different. In Stuyvesant, the creative oddballs were admired. Oddly enough, I would probably tell myself to take more time off. You're only a kid once!

5) What obstacles have you overcome as a woman in physics? How did you overcome them?

Seriously, for women I think there are obstacles every day. We are so used to them we ignore them or just suppress them a lot of the time. But I might even ask for directions or a question about physics, and the guy we talk to will inevitably answer the guy standing next to me. You have to shout a little louder to be heard. And then you are a little bit resented for doing that. There is no easy solution.

Having said that, you do good work and focus on what's important, and a lot of people are a lot more normal. But there are always people who will give you a hard time. You just have to ignore them or prove them wrong!

6) What advice would you give to aspiring women going in STEM? Anyone going into STEM?

Just do your best. Keep your eye on the prize. It's the same advice I would give to aspiring men by the way!

7) What are the biggest factors that have helped you be successful? What are your success habits?

I like to think I had some talent, and I also worked hard. I am very undisciplined, oddly enough. But when I get absorbed I like to focus a lot!

8) What are your favorite parts about being a researcher? An author? A professor?

I love problem solving. I like when people respond well to what I share with them and how I do it and when people admire my writing. But I also like the positive effects it has, such as inspiring people. Students can be amazing.

9) What are your least favorite/hardest parts about being a researcher? An author? A professor?

The problem is doing all three. I just don't have enough time. Any one of these things can take all your attention.

10) You have won numerous awards, contributed to the Randall–Sundrum model, and written multiple books that have been on the New York Times 100 Notable Books list, among several other incredible accomplishments. What would you consider to be your greatest accomplishment? Why?

Staying true to myself through it all. You have to remember what you think is important.

Stuyvesant Faculty Perspectives: Mr. Peter Brooks: A Crafter Who Crafts His Own Universes in a Machine

Debolina Kunda

Mr. Peter Brooks has taught mathematics and computer science at Stuyvesant since 2005. He received his Bachelor's degree in mathematics from California Institute of Technology, an M.A. in economics from The New School, and an M.A. in Math Education from Teachers College, Columbia University. In 1984, Mr. Brooks founded MicroMind, Inc., a software development company which he ran for 13 years. Along the way, he co-developed the first interpreter for the C programming language and later created one of the first World Wide Web browsers, for which he received a national award. He also created seven other software products for the market. In 2003, he was invited to Stuyvesant to talk to students about advanced mathematics and computer science ideas. Soon after that experience, he gave up full-time software development to get his teaching license. Since then, he has taught classes such as the annual Introduction to Computer Science, Human Reasoning, and Artificial Intelligence. (Math Survey, 2015)

1 The Road That Led to Stuyvesant

Growing up with a craftsman father made Mr. Brooks interested in envisioning an idea and recreating it, which Computer Science (CS) is all about. His father, a dollmaker, built intricate visions into toys and dolls, and Brooks has crafted his own tiny universes in the mind of a machine.

This beloved teacher did not always think that he would be entering this field. When he was in tenth or eleventh grade, his aptitude and deep enthusiasm for physics and mathematics had convinced him that it would be remarkable to become a mathematical nuclear physicist. What convinced the 16-year-old Brooks even further was the way those three words sounded. However, the change of direction occurred when he was trying out various courses and found his true love: CS, obviously!

After running his own software development company and before coming to Stuyvesant, Brooks consulted on Wall Street, allowing him to leave work at 5 PM, spend time with family, work on outside projects, and most definitely read his favorite nonfiction books. He discovered that teaching consumes a great deal of time outside of the school hours, making it infeasible to work on the significant projects that he is used to creating. However, the interaction with students and watching them incorporate their artistic abilities in CS makes teaching worthwhile for him.

2 The Joy of Teaching

To Brooks, teaching the annual Introduction to Computer Science course is a golden opportunity. He can introduce an entirely new topic to students and redirect them into CS from a completely different field. Unlike in more advanced courses like Advanced Placement CS, in intro courses, he is able to “grab [our] head[s], shake it, and make it enthusiastic!” About six years ago, Brooks had created a year-long course named

Human Reasoning, which was based on critical thinking. Students were taught about fallacies, intellectual biases (identifying and getting around them), mental calculations, estimation, how to guess, and humor. Then he became interested in teaching mechanical thinking and started teaching Artificial Intelligence.

Brooks does not want students to just “learn another [EXPLETIVE] language with its own syntax and grammar,” but to be able to use it for eye-opening artistic or scientific work. He provides students with a canvas for the competition for the best personal homepage, which ends up being a “cornucopia of creativity” since some employ their graphic design skills. Grading work is always a prickly for teachers, but the needle’s tip is sharper for CS teachers. To grade a CS exam, one has to think from the perspective of a computer and decide what mistakes the computer would have detected. Doing this for over 150 students is definitely a feat, making the Stuyvesant community appreciate Brooks and the other CS teachers greatly.

According to Brooks, the CS Department has wonderful camaraderie, where the teachers observe how each other teaches, learn each other’s materials, and develop ideas together. He describes the tight bonding as “warming to the soul.” This group has many discussions and at times, arguments, on what specific materials should be taught to the students, as there are numerous intriguing subjects in CS to choose from. Unlike teachers in the other departments in Stuyvesant, the CS Department teachers have a greater scope for developing their own curriculum. This greatly benefits students as they can learn fascinating topics that many students in other schools cannot access. When teachers can choose topics they are very interested in, students naturally absorb the enthusiasm.

A common difficulty that Stuyvesant teachers face in remote learning is not being as familiar with the students as they would have been when teaching in person. Brooks falls in this category and finds the Zoom setting not very beneficial in conversing and being acquainted with all the students. In addition, Zoom fails to allow the students to capture his amusing nature and witty humor. To compensate for the limitation of learning that students obtain virtually, Brooks thoroughly looks through every homework assignment and gives feedback through his homework server instead of just scanning through them for completion as done before the pandemic.

3 Perspectives on Computer Science

CS is a field that has been expanding our horizons and will continue to do so in the future. It will be prevalent in many areas of expertise, such as data science. According to Brooks, data science is “taking the mass of raw data from a particular application, for instance, election statistics [...] and cleaning it, so to speak, or making it uniform so a machine can actually digest it.” Then one can compare the data and merge it with other pieces of data. This leads to the next step, which is machine learning; machines will look for patterns in the data or “gauge the strength of patterns that we see.”

Another area in which CS will leave a huge footprint is cybersecurity. Brooks states that it is “a very broad area that ranges from antivirus programs to cracking cellphones to defending a company against various cybersecurity attacks.” In today’s world, it is too easy to access other people’s information, making the environment perilous. However, the significant growth of cybersecurity has been encouraging.

The STEM fields are generally known for lacking gender diversity. Women account for only 27 percent in the STEM fields and make up 20 percent of CS professionals. Our school’s statistics demonstrate the same imbalance. All students take Introduction to CS as a requirement in their sophomore year, making the sex

ratio the same as any other mandatory course, around 50 percent. However, according to Brooks' data, as we reach senior-level CS courses, the percentage of female students drops significantly to about 25 percent.

When asked to explain the reason for the imbalance, he stated that both genders have slightly different personalities, "with men being more aggressive and show-offs." However, Brooks raises a bigger question: Should we see fewer women in CS as a problem? Instead of taking two semesters of CS, many students, especially female students, take a semester of a design course to fulfill their 5-Tech and 10-Tech requirements, like Architecture or Jewelry Design, that encourages their creativity. Urging female students to take CS is equivalent to saying CS is superior to other courses, which is a matter of opinion. Thus, although Brooks believes that there are not many women in CS classes, it is not a bad situation for women to be exposed to other potential interests which would be unavailable if they were on a CS track.

4 Sage Advice for Students

Using his own life as an illustration, Brooks advises students to "find [something] amongst the things that you deeply enjoy. Try many things because you really do not know what you will enjoy. Use your college years to experiment with lots of things. You are going to have less of a chance afterward."

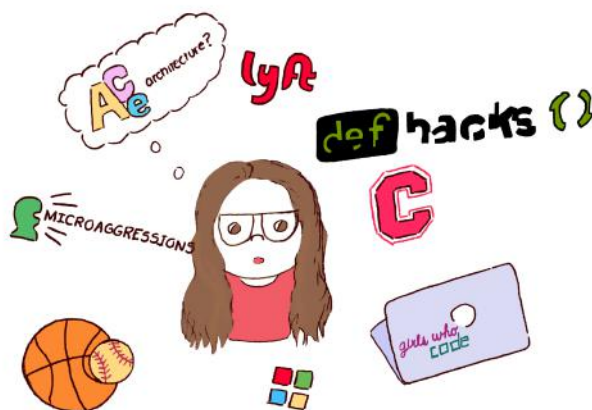
According to Brooks, the old proverb for aspiring writers, "Kill your darlings," works for computer programmers as well. Just like writers, computer programmers have to let go of the draft that took countless hours, only to start a new one. Those of us who are prepared to go through such frustration are ready to choose CS as a career path.

In our modern materialistic world, we focus on creating something to make a lot of money. However, Brooks tells us, "You are a rare individual who has the talent and the enthusiasm for something that the society really wants [...] Try to choose amongst the vast number of possible ways to spend your talents in ways that matter to society. You have that rare choice that most people do not."

Stuyvesant Alumni Perspectives: Caitlin Stanton, Class of 2016

Isabelle Lam

As a female in STEM, Caitlin Stanton (Class of 2016) shares her viewpoints on several challenges and successes during high school and college.



1 Memories of Stuy

When Caitlin Stanton first started at Stuyvesant, she truly thought she would pursue a career in architecture. To fuel this interest, she took advantage of a national career immersion opportunity named ACE (Architecture-Construction-Engineering) Mentor Program of America. Surprisingly, this program showed her that she didn't really want to study architecture after Stuyvesant. Although she stayed with the program for all four years of high school, Caitlin realized that she enjoyed only certain parts of architecture, such as the artistic and functional aspects.

When her parents enrolled her in a Girls Who Code (GWC) program, Caitlin found a supportive environment to learn about STEM. Although she initially wasn't excited about attending this seven-week program, she discovered that she enjoyed learning in a STEM environment. In her Stuyvesant career, she initially didn't enjoy Introduction to Computer Science, since the processes in Computer Science (CS) are much different from other classes. She recalled, "Debugging my final project [...] was a process that I hadn't been used to before. Most other work you do in high school was kinda just cut-and-dried, like you read a history

textbook or solve a math equation. There's not a lot that you can fix, not a lot of problems you can solve. Code is a whole different story." Instead of just memorizing, regurgitating, and then promptly forgetting what she learned, CS challenged her to really understand her code. It was a "really different experience from the rest of high school," she said.

Despite this newfound interest in CS, Caitlin pursued other extracurriculars while attending Stuyvesant. She was one of the co-captains of the softball team, participated in ARISTA, and was the basketball team manager. She explained, "I knew that I liked volunteering and pushing myself academically, so ARISTA made a lot of sense. I also knew I enjoyed sports like basketball and softball. It was nice to have a chunk of my time put aside for things that I knew I enjoyed, outside of going home and doing homework." She advised against padding college applications by joining a lot of clubs, especially if you don't have any interest in those clubs. She noted how a lot of her peers at Stuy exaggerated their involvement in clubs without contributing much and joined just for the sake of listing 15 clubs on their resume. She remarked, "It's not necessarily worth spending your time doing things that you don't enjoy just to get into a certain college. You don't want to look back on high school and think, 'Wow, I spent my time doing things I didn't like, all for nothing.'"

In her senior year, she co-founded a hackathon named def hacks() with her friend Emily Redler. After observing a "I don't belong here" mentality coupled with being a "16 or 17 year old girl" at traditional hackathons, Caitlin realized that she should create her own. This new hackathon would not be limited to students in specialized high schools and would not instill a sense of inferiority, one of her struggles. She noted, "I wanted to create a space for high school students, by high school students." At first, it was difficult getting sponsorships for venues and spreading the word about this new organization. Caitlin explained, "We had a bunch of people we could reach out to. All we really had to do was prove to them that our concept was worth it." Many times companies didn't want to donate to def hacks() because they were just highschoolers, and they didn't have any other sponsors. But after the first company, ThoughtWorks, made a contribution, other companies' donations began trickling in quite easily. Despite how arduous this experience was, Caitlin found it very rewarding. Both she and Emily were able to gain a new skillset and many new experiences. For example, some companies were hesitant to provide large sums of money, but Caitlin was able to negotiate smaller sums, preventing a net loss for her hackathon. After the first event, companies were more open to sponsoring, since she was able to show substantial participation in her hackathon.

2 Majoring in CS in College, Confronting Bias

In college life at Cornell University, Caitlin has encountered biases against women within her major, Electrical and Computer Engineering (ECE). She explained how she's "lucky enough to never have been outright discriminated against," but, "what's more of the norm is unconscious bias or microaggressions. A lot of my male friends will not turn as quickly to me for homework help, they'll interrupt me while I'm talking, or they'll restate the ideas that I said five minutes ago." She elaborated, stating, "It also shows how maybe that person didn't mean anything by that comment, but it came so easily to them to say." She recognizes that "most of the time, these are my friends. I know that they don't actively mean anything by it, it's just the way that society has raised them. The environment they were put in places women so they feel they can't be a part of the STEM community because they can't have more 'masculine' traits or personalities." She proposes teaching girls from a young age about their self-worth and providing adequate and easily accessible resources and opportunities for girls.

Although Cornell boasts a 50/50 female to male ratio, that ratio only applied to beginner STEM courses.

Caitlin noticed the ratio becoming smaller and smaller as she took more advanced courses. To bridge the gap, some of her professors have taken steps to try to solve this problem. She reminisced, “Last year, I had a professor for a notoriously difficult CS course who said at one of the first lectures, ‘Hey everyone, you belong here. You belong in this class.’ [...] Having someone who I didn’t know, and who is a white, cis-man, tell me, ‘Hey, this is a space for you. I actively want to make this inclusive and I actively want to help everyone think that this is a class that they can take and succeed in.’ That meant a lot.”

Other teachers were uplifting in more quiet ways. Her Fast Robots professor, a female, noticed that there was only one woman enrolled in the course, and emailed all of the women of the ECE Department with details about the course. Despite this thoughtful gesture on the professor’s part, Caitlin lamented that “the fact that I had to have a professor reach out to me to find a certain class and to enroll in it is kinda sad.” Due to the professor’s efforts, the ratio of women to men in the class trended upwards and was 1:3 at the time of interview.

One professor took a non-confrontational approach. Caitlin recalled how a male student in her class who outright refused to ask female teaching assistants (TAs) for help. The professor didn’t correct the male student, but gave TAs permission to kick someone out of office hours for rudeness, which felt empowering. Caitlin also noted how some females in her engineering sorority did not have a single female engineering professor at Cornell. In contrast, Caitlin has had two so far.

As for Cornell’s overall effort to eliminate bias, she commented, “I think Cornell is taking strides. I don’t think they’re big enough yet though. There’s a long way to go for them to be able to say that the college in general is an equitable place for people of any gender and people of any identity.”



3 The Working World and Beyond

Outside of academic life, Caitlin enjoyed largely positive internships at technology titans such as Lyft and Microsoft where she benefited from supportive and instructive environments. In the summer of 2019, she worked as an electrical engineering intern for the first time at Microsoft. In this new environment, she was expected to troubleshoot Bluetooth and WiFi for the Microsoft Surface laptops. At Lyft, where Caitlin interned virtually in June and July 2020, it was her first time combining machine learning (ML) with

hardware. Before this internship, she had thought of ML as a software concept and not implementation into hardware. She was tasked with researching ML chips for the car and looking for potential partner companies. Although she was given a large amount of responsibility in making a chip recommendation, she felt that the people at Lyft were very receptive and willing to help. This would be an ideal environment for younger girls to nurture interests in STEM.

Caitlin is currently pursuing a Masters of Engineering at Cornell, a one-year program that is more focused on a project than on research. She describes ECE as “[doing] a great job of illustrating the wide range of field from lowest level hardware in silicon chip design to highest level software in user-facing apps.” After she earns her master’s, she hopes to go into a Hardware and Engineering role, where she can build circuits and code up hardware. For circuits, she plans to intern at Samsung Research America (S.R.A.) this coming summer, where she will be working with emerging technology as part of the Think Tank Team. Hardware is the branch of CS that focuses on “the process of designing, developing, testing and producing computer systems and various physical components related to computer systems.” She thinks companies such as Apple, Samsung, and Google are always looking for hardware engineers with innovative ideas. However, she doesn’t want to limit herself to technical pursuits. She plans to go into public speaking and continue advocating for STEM because she really enjoys sharing her experiences with larger audiences. Long term, Caitlin plans to explore entrepreneurship in business, marketing, and sustainability as her way to make a mark on the world. To do this, she envisions becoming an executive manager or board member so she can work with groups of people, share ideas, and keep her career both technical and non-technical.

4 Statistics and References

The national average female to male ratio in STEM courses is 17%. The Society of Women Engineers provides more in-depth statistics, reporting that “over 32% of women switch out of STEM degrees in college” with “only 30% of the [68% or less of women] who earn bachelor’s degrees in engineering still working in engineering 20 years later.”

According to Cornell, the amount of women in ECE has increased to 35% in 2017 from 13% in 2012. They also note that “for the first time in Cornell Engineering’s history, the college has admitted a class composed of more women than men, with the Class of 2021 being 51% female.” These statistics fall within the gender binary and do not cover the full gender spectrum as accepted today.

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Mathematics for Human Flourishing

Distinguished Guest Contributor
Professor Francis Su

*Why does the practice of mathematics often fall short of our ideals and hopes? How can the deeply human themes that drive us to do mathematics be channeled to build a more beautiful and just world in which all can truly flourish? Professor Su gave this Retiring Presidential Address at the American Math Society-Mathematical Association of America Joint Winter Meetings on January 6, 2017 at the end of his term as President of the Mathematical Association of America (MAA). This speech was published in the June 2017 issue of The American Mathematical Monthly, and a full-length book building on this speech, but aimed at the general public, was published by Yale University Press in 2020: bit.ly/math-flourish-book. *Mathematics for Human Flourishing* won the Mathematics Association of America's 2021 Euler Book Prize. The MAA and the American Mathematical Society also jointly published a book, *Living Proof: Stories of Resilience Along the Mathematical Journey*, which can be found at www.maa.org/livingproof. Math Survey obtained permission from Professor Su to republish this article for the 2020-2021 Winter Edition.*

Every being cries out silently to be read differently.
— Simone Weil, *Gravity and Grace*

Christopher is an inmate in a high-security federal prison not far from Atlanta. He's been in trouble with the law since he was 14. He didn't finish high school, had an addiction to hard drugs, and at age 21, his involvement in a string of armed robberies landed him in prison with a 32-year sentence.

Right now, you've probably formed a mental image of who Christopher is, and you might be wondering why I'm opening my speech with his story. When you think about who does mathematics—both who is capable of doing mathematics and who wants to do mathematics—would you think of Christopher?

And yet he wrote me a letter after 7 years in prison. He said: "I've always had a proclivity for mathematics, but being in a very early stage of youth and also living in some adverse circumstances, I never came to understand the true meaning and benefit of pursuing an education... over the last 3 years I have purchased and studied a multitude of books to give me a profound and concrete understanding of Algebra I, Algebra II, College Algebra, Geometry, Trigonometry, Calculus I and Calculus II." Christopher was writing me for help in furthering his mathematics education.

When you think of who does mathematics, do you think of Christopher?

Every being cries out silently to be read differently.



Simone Weil is a well-known French religious mystic and a widely revered philosopher. She is probably less well known as the younger sister of Andre Weil, one of history's most famous number theorists.

For Simone Weil, to 'read' someone means to interpret or make a judgment about them. She's saying: every being cries out silently to be judged differently. I sometimes wonder if Simone was crying out about herself. For Simone, too, loved and participated in mathematics, but she was always comparing herself to her brother. She writes:

"At fourteen I fell into one of those fits of bottomless despair that come with adolescence, and I seriously thought of dying because of the mediocrity of my natural faculties... the exceptional gifts of my brother, who had a childhood and youth comparable to those of Pascal, brought my own inferiority home to me. I did not mind having no visible successes, but what did grieve me was the idea of being excluded from that transcendent kingdom to which only the truly great have access and wherein truth abides. I preferred to die rather than live without that truth."

We know Simone loved mathematics because she used mathematical examples in her philosophical writing. And you'll find her in photos of Bourbaki with her brother.



(Bourbaki, 1937)

I often wonder what her relationship to mathematics would be like if she weren't always in Andre's

shadow.

Every being cries out silently to be read differently.

As MAA President, you might think that my relationship to mathematics has always been solid. I don't like the word 'success' but people look at me and think I'm successful, as if the true measure of mathematical achievement is the grants I've received, or the numerous papers I've published.

Like Christopher, I've had a proclivity for mathematics since youth.

But I grew up in a small rural town in South Texas, with limited opportunities. Most of my high school peers didn't even attend college. I did because my dad was a college professor, but my parents didn't know about the many mathematical opportunities I now know exist.

My love for math deepened at the University of Texas, and I managed to get admitted to Harvard for my PhD. But I felt out of place there, since I did not come from an Ivy League school, and unlike my peers, I did not have a full slate of graduate courses when I entered. I felt like Simone Weil, standing next to future Andre Weils, thinking I would never be able to flourish in mathematics if I were not like them.

I was told by one professor that I didn't belong in graduate school. That forced me to consider, among other things, why I wanted to do mathematics. And in fact, that is essentially the one big question that I'd like for you to consider today:

Why do mathematics?

This is a simple question, but worth considerable reflection. Because how you answer will strongly determine who you think should be doing mathematics, and how you will teach it.

Why is Christopher sitting in a prison cell studying calculus, even though he won't be using it as a free man for another 25 years? Why was Simone so captivated by transcendent mathematical truths? Why should anyone persist in doing math or seeing herself as a mathematical person when others are telling her in subtle and not so subtle ways she doesn't belong?

And in this present moment, the world is also asking what its relationship with mathematics should be. Amidst the great societal shifts wrought by the digital revolution and a shift to an information economy, we are witnessing the rapid transformation of the ways we work and live. And yet we hear voices in the public sphere saying: "high school students don't need geometry" or "let's leave advanced math for the mathematicians". And some mathematicians won't admit it but they signal the exact same thing by refusing to teach lower-level math courses, or viewing the math major as a means to weed out those they don't think are fit for graduate school.

Our profession is threatened by voices like these from within, and without, who are undermining how society views mathematics and mathematicians. And the view of our profession is dismal. The 2012 report from the President's Council of Advisors on Science and Technology pegs introductory math courses as the major obstacle keeping students from pursuing STEM majors. We are not educating our students as well as we should, and like most injustices, this hurts those who are most vulnerable.

I want us as a mathematical community to move forward in a different way. It may require us to change our view of who should be doing mathematics, and how we should teach it. But this way will be no less rigorous and no less demanding of our students. And yet it will draw more people into mathematics because they will see how mathematics connects to their deepest human desires.

So if you asked me: why do mathematics? I would say: mathematics helps people flourish. *Mathematics is for human flourishing.*

The well-lived life is a life of *human flourishing*. The ancient Greeks had a word for human flourishing, *eudaimonia*, which they view as the good composed of all goods. There is a similar word in Hebrew: *shalom*, which is used as a greeting. Shalom is sometimes translated ‘peace’, but the word has a far richer context. To say ‘shalom’ to someone is to wish that they would flourish and live well. And Arabic has a related word: *salaam*.

A basic question, taken up by Aristotle, is: how do you achieve human flourishing? What is the well-lived life? Aristotle would say flourishing comes through the exercise of virtue. The Greek concept of virtue is “excellence of character that leads to excellence of conduct.” So it includes more than just moral virtue; for instance, courage and wisdom are also virtues.

What I hope to convince you of today is that the practice of mathematics cultivates virtues that help people flourish. These virtues serve you well no matter what profession you choose. And the movement towards virtue happens through basic human desires.

I want to talk about five desires we all have. The first of these is play.

1. PLAY

It is a happy talent to know how to play.
— Ralph Waldo Emerson

Think of how babies play. Play is hard to define, but since we’re mathematicians, we should at least try. We can think of a few qualities that characterize it. For instance, play should be *fun* and *voluntary*, or it wouldn’t be play. There is usually some *structure*: even babies know ‘peekaboo’ follows a certain pattern, but there is lots of *freedom* within that structure. That freedom leads to *investigation* of some sort, like ‘where will you appear if we play peekaboo one more time?’ There is usually *no great stake* in the outcome. And the investigation can often lead to some sort of *surprise*, like appearing in a different place in peekaboo. Of course, animals play too, but what characterizes human play is the enlarged role of mind and the *imagination*.

Think about Rubik’s cube or the game Set. There’s interplay between structure and freedom, and no great stake in the outcome, but there’s investigation that can lead to the delight of solving the cube or finding sets of matching cards.

Mathematics makes the mind its playground. We play with patterns, and within the structure of certain axioms, we exercise freedom in exploring their consequences, joyful at any truths we find. We even have a whole area of mathematics known as ‘recreational mathematics’! Do you know another discipline that has

a ‘recreational’ subfield? Is there a ‘recreational physics’ or ‘recreational philosophy’?

And mathematical play builds virtues that enable us to flourish in every area of our lives. For instance, math play builds **hopefulness**: when you sit with a puzzle long enough you are exercising hope that you will eventually solve it. Math play builds **community**—when you share in the delight of working on a problem with another human being. And math play builds **perseverance**—just as weekly soccer practices build up the muscles that make us stronger for the next game, weekly math investigations make us more fit for the next problem whatever that is, even if we don’t solve the current problem. It’s why the MAA supports competition programs like the AMC and Putnam. We help kids flourish through building hopefulness, perseverance, and community. This year, you may have heard that the US team, that MAA trained, won the International Math Olympiad for the second time in a row. What you might not have heard is that Po Shen Loh, who coached our team, invited teams from other countries to train with them to prepare for the IMO. You see, our priority was community over competition. This was so impressive to the Singaporean prime minister that he publicly thanked Obama for this remarkable collaboration. This was true play: teams in friendly competition.

Play is part of human flourishing. *You cannot flourish without play.*

And if mathematics is for human flourishing, we should ‘play up’ the role of play in how we teach and who we teach. Everyone can play. Everyone enjoys play. Everyone can have a meaningful experience in mathematical play.

And teaching play is hard work! It’s actually harder than lecturing, because you have to be ready for almost anything to happen in the classroom, but it’s also more fun. Play is part of what makes inquiry-based learning and other forms of active learning so effective. There’s overwhelming evidence that students learn better with active learning.

This year, in the Conference Board of the Mathematical Sciences, I signed a statement with presidents of other math organizations endorsing active learning (and you can find this on the CBMS website). And if you want to see the evidence for active learning, we’ve included some background information in this statement.

So: teach play.

Another basic human desire is:

2. BEAUTY

It is impossible to be a mathematician without being a poet in soul.

— Sofia Kovalevskaya

Who among us does not enjoy beautiful things? A beautiful sunset. A sublime sonata. A profound poem. An elegant proof.

Mathematicians and scientists are awed by the simplicity, regularity, and order of the laws of the universe. These are called “beautiful”. They feel transcendent. Why should mathematics be as powerful as it is? This is what Nobel prizewinning physicist Eugene Wigner called “the unreasonable effectiveness of mathematics”

to explain the natural sciences. And Einstein asked: “How can it be that mathematics, being after all a product of human thought independent of experience, is so admirably adapted to the objects of reality?”

And mathematicians are not satisfied with just any proof of a theorem. We often look for the best proofs, the simplest or most pleasing. Mathematicians have a special word for this: we say a proof is “elegant”. Paul Erdős often spoke of “The Book” that God keeps, in which all the most elegant proofs of theorems are kept.

Pursuing mathematics in this way cultivates the virtues of **transcendence** and **joy**. By joy, I refer to the wonder or awe or delight in the beauty of the created order. By transcendence, I mean the ability to embrace mystery of it all. There’s a transcendent joy in experiencing the beauty of mathematics.

If mathematics is for human flourishing, we must help others see its beauty.

But there are many notions of beauty. So the way you motivate mathematics through beauty must necessarily be diverse—through art, through music, through patterns, through rigorous arguments, through the elegance of simple but profound ideas, through the wondrous applicability of these ideas to the real world in many different fields.

A third basic human desire is:

3. TRUTH

Quid est veritas? (What is truth?)

— Pontius Pilate

What is truth? This question is an important one, especially today. Each day seems to bring more discussion about how fake news may have influenced the presidential election. Some dismiss trying to figure out what’s true, saying ‘How can we even know what is true?’

And yet, in some contexts, people seek truth at all costs. Especially when there is a lot at stake. When my dad had cancer, we wanted to know what treatments had the best chance of saving his life. We had to know, because his life depended on it.

The quest for truth is at the heart of the scientific enterprise. I say ‘quest’ because we don’t do science to confirm simple declarative statements that are easily verified, like: “my coffee is hot”. Rather, our subject of investigation are questions for which the answer is not so clear. “Do gravity waves exist, and if so how would we detect them?”

So there is a quest. We formulate a hypothesis (“they exist”) and we design experiments to test our hypotheses. We look for evidence, and if we find some, we still ask: could there have been any other explanation?

A mathematician might try to prove or disprove a statement through logical deductions from first principles. Or she might construct a mathematical model to answer the question.

These approaches cultivate in us the virtue of **rigorous thinking**: the ability to handle ideas well, and craft clear arguments with those ideas. This virtue serves us well in every area of life. We should use this ability to reason in the public square, as many in our community have done by writing op-ed pieces for newspapers.

I would love to see more of us exercise this virtue, to shape public perceptions about mathematics.

(And, on a side note, if you want to see how math is being portrayed in the news, I have built a ‘math news aggregator’ called *MathFeed*, which is available for iPhone and iPad, and in a few weeks, Android. It links to major newspapers, as well as math blogs and publications of MAA, AMS, SIAM, ASA and other organizations.)

I would like to encourage institutions to start valuing the public writing of its faculty. More people will read these pieces than will ever read any of our research papers. Public writing is scholarly activity: it involves rigorous arguments, is subject to review process by editors, and to borrow the NSF phrase, it has broader impacts, and that impact can be measured in the digital age.

And, the quest for truth predisposes the heart to the virtue of **humility**.

Isaac Newton said: “I do not know what I may appear to the world, but to myself I seem to have been only like a boy playing on the seashore, and diverting myself in now and then finding a smoother pebble or a prettier shell than ordinary, whilst the great ocean of truth lay all undiscovered before me.”

He’s saying: the more we know, the more we realize how much we do not yet know. And we learn how to accept being wrong, if a counterexample shows our conjecture was false. In fact, I’ll go so far to say that counterexamples in mathematics have a special place— we celebrate them. We have titles of books like ‘Counterexamples in Topology’ or ‘Counterexamples in Analysis’. We like to admit when we are wrong!

So when a student embraces this quest for truth, she begins to assume a certain kind of humility. She handles ideas rigorously, with honesty and integrity. She values truthfulness and clarification of distinctions. This is the virtue of intellectual humility, and it is prized. I think a lack of humility characterized the political discourse of 2016 on both sides. I wish we had more intellectual humility in the public square.

We must model the virtue of humility in our own teaching, and we should explicitly tell our students we are cultivating humility as a virtue that will serve them well their entire life. One of the most important skills we can teach our students is to know when their arguments are wrong. How many of you have ever given a super-hard question on an exam, and gotten answers that look like students just made stuff up, hoping for extra credit? I now explicitly say on my exams: that I will give extra credit on incomplete proofs where students acknowledge their gaps. I get much more thoughtful answers that way.

And mathematics builds the virtue of **circumspection**. We know the limits of our arguments and we don’t overgeneralize. I like what my friend Rachel Schwell said:

“I think math helps me make fewer sweeping generalizations about people. For example, I wouldn’t assume a person is say, uneducated, just because she is, say, poor, just as I can’t assume a number is say, positive, because it is an integer. I can’t even assume it is positive if I know it’s nonnegative, even if,

probability-wise, it probably is positive! So I don't leap to automatic associations as much."

Can we help our students see that the virtue of circumspection is important in life?

A fourth basic human desire is:

4. JUSTICE

Justice. To be ever ready to admit that another person is something quite different from what we read when he is there (or when we think about him). Or rather, to read in him that he is certainly something different, perhaps something completely different from what we read in him. Every being cries out silently to be read differently.

— Simone Weil

Akemi was a student of mine who did research with me as an undergraduate. Her innovative paper linking game theory and phylogenetics was published in a highly regarded mathematical biology journal. She went to a top research university for her PhD. So I was surprised when I learned that Akemi quit after one year.

She told me she had many negative experiences. Her advisor was never willing to meet with her, and she had faced uncomfortable experiences as a woman. She told me one example:

"At the beginning of the course, I consistently got 10/10 on my homework assignments which were all graded by the TA. One day, Jeff [a mutual friend] told me that he was hanging out with our TA and someone asked the TA how the analysis class was doing. He went on and on about some "guy" named Akemi and how perfect "his" homeworks were and how clearly they were written, etc. Jeff told him I was a girl and the TA was shocked. (Jeff told me this story because he thought it was funny that someone both didn't know my sex from my name and reacted so dramatically to finding out.) After that, I never got remotely close to 10/10 on my assignments and my exams were equally harsh—most of the reasons for docked points were vague with comments like 'give more detail'. I didn't feel like my understanding of the material diminished that quickly or dramatically, but I suppose it's possible that happened and I'm just misinterpreting the situation..."

No, Akemi, it's not you.

I hope you all agree something is not right with this picture. If a certain kind of anger wells up in you, you are experiencing a telltale sign of flourishing: the desire for justice. Justice means: setting things right. And justice is a powerful motivator to action.

Justice is required for human flourishing. We flourish—we experience shalom—when we treat others justly and when we are treated justly.

As the quote above reveals, Simone Weil realized that correcting injustice must involve changing how we view others. She defines: "Justice. To be ever ready to admit that another person is something quite different from what we read when he is there (or when we think about him). Or rather, to read in him that he is certainly something different, perhaps something completely different from what we read in him. Every

being cries out silently to be read differently.”

Now before we are too quick to censure Akemi’s TA, we have to realize the problem of reading others differently begins with ourselves. The TA may not have even realized he was doing this. This is the problem of *implicit* bias: unconscious stereotypes that subtly affect our decisions. One of the best experiences I had in MAA leadership was attending a workshop on implicit bias, in which I realized in a powerful way how I am biased even though I try not to be. We all do it without realizing it. Numerous experiments confirm results of the following kind: when given two nearly identical resumes except that one has a positively stereotyped name and one has a negatively stereotyped name (women, minority, etc), judges will rate the positively stereotyped resume higher. This happens even if judges come from the negatively stereotyped group.

This is why good practices are important. The MAA now has a document for selection committees called ‘Avoiding Implicit Bias’ that lists a number of practices that have been shown in research to mitigate the effects of implicit bias, such as taking time to make decisions, or generating a large candidate pool. These are good practices, even if you don’t believe bias exists. That document is now distributed with every MAA committee assignment.

You see, we have to recognize that even if people are just, even if they desire to be just, a society may not be just if its structures and practices are not also just. And the only way a whole society can flourish, is if the society is a just society. It is often said that the mark of a just society is how it treats its most vulnerable members.

So I ask, with great humility: are we a just community?

If you believe that mathematics is for human flourishing, and we teach mathematics to help them flourish, you will see, if you look around the room, that we aren’t helping all our students flourish. The demographics of the mathematical community does not look like the demographics of America. We have left whole segments out of the benefits of the flourishing available in our profession.

So we have to talk about race and that’s hard. It can bring up complicated emotions, even more so with all that has taken place in our nation in the past year. In our community, we have to become more comfortable talking about it, listening to each others’ experiences, and being willing to recognize its there. If you want to treat others with dignity and they are hurting, you don’t ignore their pain. You ask “What are you going through?”

It’s not enough to say “I don’t think about race.” Because in a community, how one member is doing affects the whole. And for those of us not in the dominant racial group, we don’t have the luxury of saying “I don’t think about race” because racial issues affect us on a daily basis. So let me encourage all of us to try having these conversations, to be quick to listen, slow to speak, and quick to forgive each other when we say something stupid. That’ll happen if you start to have conversations, and we just have to have grace for each other if we make mistakes—it’s better than not talking.

So if we’re going to have conversation, I’ll start. I grew up in Texas in a white and Latino part of the state, and I realized early on that my family had different customs from my friends—my clothes were different, the food in my lunchbox was different—and these things were causing me to not fit in. I wanted to be white. Not Latino, because white people got more respect, and as an Asian I was getting picked on

all the time. I had no role models for being Asian-American. So I tried hard to act white, even if I couldn't look white.

On the other hand, in Chinese communities, I also don't fit in. I don't speak Chinese. I don't act Chinese. At Chinese restaurants, I'm viewed as white. Did you know at authentic Chinese restaurants, there is often a special menu, a secret menu, that they only give Chinese people? It has all the good stuff. I don't get that menu, unless I ask for it. In fact they discourage me: saying 'you won't like the stuff on that menu'

As mathematicians, who gets to see our secret menu? Whom do we shepherd towards taking more math courses? Whom do we discourage from looking at that menu?

Don't let me sound as if I'm complaining about my race. There are ways in which I benefited from being Asian. People expected me to do better at math and science, and I'm sure that's part of why I did. Because I now know there is a recognized literature on 'expectancy effects', that teacher expectations do affect student performance.

The first time I didn't feel like a minority was when I moved to California. There are so many Asian-Americans there. In Texas, I would commonly get the question "Your English is so good! Where are you from?" "Texas." "No, where are you really from?" That never happens in California and there's a feeling of freedom I have in not having to counter these verbal stings.

These days, I'm used to being at math conferences and seeing a sea of white faces. So even I was a little bit surprised, that when I was elected MAA President, a prominent blogger on race issues for Asian Americans wrote a blog post about it. His name is Angry Asian Man. He looked at the photos of past MAA Presidents on our website, and given how many Asians he expected to be in math, he noted they were all white except for me, and wrote a sarcastic post entitled:

"Finally! An Asian guy who's good at math!"

I am the first president of color of either AMS or MAA. Minorities, including Asians, are easy to overlook when you think about who would make a good leader. This may not be intentional, but when you are asked to think about who is fit for this or that role, you often think of people just like the people who have been in office. So it is easy for implicit bias to creep in.

I raise this discussion out of deep affection for the mathematical community. I want us to flourish, and there are ways in which we can do better.

In 2015, I had the great pleasure of running MSRI-UP, a summer research program for students from underrepresented backgrounds: first-generation college students, Latino and African American kids. I asked them to help me prepare this talk, to tell me about obstacles they've faced doing mathematics.

One of them, who did wonderful work that summer, told me about her experience in an analysis course after she got back. She said "...even though the class was really hard, it was more difficult to receive the humiliations of the professor. He made us feel that we were not good enough to study math and he even told us to change to another 'easier' profession." As a result of this and other experiences, she switched her major to engineering.

Let me be clear: there is no good reason to tell a student she doesn't belong in math. That's the student's decision. Not yours. You see the snapshot of her progress, but you don't see her trajectory. You can't know how she will grow and flourish in the future. But you can help her get there.

Of course, you should give forthright counsel to students about skills they might need to develop further if they want to go on in mathematics, but if you see mathematics as a means to help them flourish, why wouldn't you encourage them to take more math?

Another student that summer, Oscar, told me about his experience as a math major who, unlike his peers and because of his background, did not enter college with any advance placement credit. He says:

"I noticed how different my trajectory was, however, while I was in my Complex Analysis course. A student was presenting a solution on the board which required a bit of a complicated derivation halfway through. They skipped over a number of steps, citing 'I don't think I need to go through the algebra. . . we all tested out of Calculus here anyway!' with my professor nodding in agreement and some students laughing. I quietly commented that Calculus was my first course here. My professor was genuinely surprised and said 'Wow, I did not know that! That's interesting.' I was not sure whether to feel proud or embarrassed by the fact that I was not the 'typical math student' that was successful from the beginning of their mathematical career. I felt a sense of pride in knowing that I was pursuing a math degree despite my starting point, but I could not help but feel as though I did not belong in that classroom to begin with."

The reason Oscar was in that class to begin with was because of the active support of another professor. Says Oscar:

"She presented me with my first research opportunity and always encouraged me to study higher math. I was also able to confide in her about a lot of the internal struggles I had with being a minority in mathematics since, as a female, she had a similar experience herself! My complex analysis professor became one of my mentors as well. I think it was just an interesting moment because she didn't realize how her reaction to the situation could have hurt me (and I don't think she's necessarily at fault!). It was more that her reaction piled onto the insecurities I held in regards to being a minority with a weak background in math."

(Note that Oscar didn't have a 'weak' background—he had a standard background.)

I'm pleased to say that Oscar and his team from that summer just published a paper in an AMS journal, and he is now in graduate school.

You hear from Oscar's story the importance of having an advocate, a faculty member who says "I see you, and I think you have a future in math." This can be especially important for underrepresented groups and women who already have so many voices telling them they can't. Can you be that advocate?

And if we teach mathematics to help our students flourish, then we should not set up structures that disadvantage smart students with weaker backgrounds, or make them feel 'out of place'. I know that can happen inadvertently among students, but we, as faculty, are the shepherds of our departmental culture.

When I was a grad student at Harvard, they had a regular calculus class, an honors calculus class called Math 25, and on top of that, for those with very strong backgrounds, a super honors class called Math 55.

Ironically, I regularly encountered students in the honors track who felt they didn't belong in math, because they hadn't placed into the super honors track. I had to keep reassuring them "background is not the same as ability." I sometimes wish graduate school admissions would remember this too: "background is not the same as ability". As my friend Bill Velez says: if you want your PhD program to have more students of color, then you have to stop admitting students on the basis of background, and start admitting students by their ability. And then, support them. Be an advocate! They have so many voices telling them, they don't belong.

I know our community wants to be just. To set things right. If you were at the Hidden Figures panel Thursday, you saw the turnout. There were so many people we had to open a second room and even then, it was standing room only. So I know our community wants to do better.

So if you are looking to start some conversations with your students or colleagues, but don't know where to start, sometimes it can help to have a third party. I'm willing to be that third party. I've written a number of articles on these topics for MAA FOCUS and they are all posted on my webpage. You could ask your students to read them and then have a discussion. I can assure you it will be time well spent.

Because we are not mathematical machines. We live, we breathe, we feel, we bleed. If your students are struggling, and you don't acknowledge it, their education becomes disconnected and irrelevant. Why should anyone care about mathematics if it doesn't connect deeply to some human desire: to play, seek truth, pursue beauty, fight for justice? You can be that connection.

So let me challenge each one of you today. Find one student whom you know is facing some challenges, and become their long-term advocate. One way to do that is to sign up to be a mentor with the Math Alliance. The goal of this program, directed by Phil Kutzko, is to ensure that every underrepresented or underserved American student with the talent and the ambition has the opportunity to earn a doctoral degree in a mathematical science.

Find one student and be their advocate! Be the one who says "I see you, and I think you have a future in math." Be the one who searches out opportunities for them. Be the one who pulls them towards virtue. Be the one who calls them up when they've skipped class, and asks "is everything okay? what are you going through?"

I know what I'm asking you to do is hard and takes time.

But we're mathematicians... we know how to tackle hard problems. We have the perseverance to see it through. We have the humility to admit when we make mistakes, and learn from them. We have hopefulness that our labor is never in vain and that our work will bear fruit in the flourishing of our students.

Because what I am asking you to do is something you already know, at the heart of the teacher-student relationship, pulls us towards virtue. I'm asking you to love.

5. LOVE

If I speak in the tongues of men and of angels, but have not love, I am a noisy gong or a clanging cymbal.
— Paul the Apostle

Love is the greatest human desire. And to love and be loved is a supreme mark of human flourishing. For it serves the other desires—play, truth, beauty, and justice—and it is served by them.

Every being cries out silently to be read differently. Every being cries out silently to be loved. Christopher, in prison, wasn't looking only for mathematical advice. He was looking for connection, someone to reach out to him in his mathematical space and say "I see you, and I share the same transcendent passion for math that you do, and you belong here."

When I was in the depths of despair in graduate school, struggling over many non-academic things with a professor who had said I don't belong, already interviewing for jobs because I was sure I was going to quit, one professor reached out to me, became my advocate. And he said, "I would rather see you work with me, than quit." So I stand before you now to ask you: find a struggling student, love them, be their advocate!

I'll close with this reflection by Simone Weil, as I think it gathers all the strands of my talk. After wrestling with her own insecurity in mathematics, she saw that there was a path to virtue through her struggle and that her struggle could help others. She wrote this:

"The love of our neighbour in all its fullness simply means being able to say to him: 'What are you going through?' It is a recognition that the sufferer exists, not only as a unit in a collection, or a specimen from the social category labelled 'unfortunate', but as a man, exactly like us, who was one day stamped with a special mark by affliction. For this reason it is enough, but it is indispensable, to know how to look at him in a certain way.

"This way of looking is first of all attentive. The soul empties itself of all its own contents in order to receive into itself the being it is looking at, just as he is, in all his truth. Only he who is capable of attention can do this.

"So it comes about that, paradoxical as it may seem, a Latin prose or a geometry problem, even though they are done wrong, may be of great service one day, provided we devote the right kind of effort to them.

"Should the occasion arise, they can one day make us better able to give someone in affliction exactly the help required to save him, at the supreme moment of his need.

"For an adolescent, capable of grasping this truth and generous enough to desire this fruit above all others, studies could have their fullest spiritual effect, quite apart from any particular religious belief.

"Academic work is one of those fields which contain a pearl so precious that it is worthwhile to sell all our possessions, keeping nothing for ourselves, in order to be able to acquire it."

She had found a path through struggle to virtue.

She understood that mathematics is for human flourishing.

The mathematical experience cannot be separated from love!

The love between friends who play with a mathematical problem.

The love between teacher and student working to help each other flourish.

The love of a community like the Mathematical Association of America working with each other towards a common goal: through the knowledge and virtues wrought by mathematics, to help everyone flourish.

Thank you for the opportunity to serve you these last two years. Shalom and salaam, my friends. Grace and peace to you. May you and all your students flourish.

*A full-length book, *Mathematics for Human Flourishing*, with reflections by Christopher Jackson, is available from Yale University Press (2020): bit.ly/math-flourish-book. Learn more at: <https://www.francissu.com/flourishing>.*

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